

Cross-Modal Affective Coherence: A Multimodal Framework for Detecting Discrepancies Between Externally Expressed and Internally Experienced Emotion Using EEG, Audio and Video

by

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Declaration

It is hereby declared that

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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Approval

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Abstract

Multimodal emotion recognition research has predominantly framed the problem as one of classification accuracy, seeking to identify a single ground-truth emotion label from complementary behavioural signals. This thesis reframes the problem: when an individual's external behavioural signals (audio and video) and their internal neurophysiological signal (EEG) disagree, that disagreement is itself a clinically meaningful quantity, associated in the literature with emotion suppression, masked affect, alexithymia, and affect dysregulation disorders. We term this phenomenon *cross-modal affective coherence* and develop a computational framework to detect and characterise it.

The framework is evaluated on the EAV dataset (Lee et al., 2024), a cue-based laboratory corpus of 42 subjects with synchronised 30-channel EEG, audio, and video recordings across five emotion categories. Per-modality encoders — the Audio Spectrogram Transformer (AST), a facial-emotion Vision Transformer (ViT), and EEGNet — are fine-tuned per subject. A trimodal cross-modal attention fusion module with 2.28 million parameters and a self-attention architecture over a three-token modality sequence is then trained on cached per-modality features.

Across 42 subjects, the per-modality mean accuracies are: audio 57.1%, vision 74.6%, and EEG 43.9%, all substantially exceeding the EAV paper's published baselines of 36.7%, 52.8%, and 36.7% respectively. Multimodal fusion yields 80.2% (cross-attention) and 80.0% (naïve late fusion), a gain of 5.4 percentage points over the best single modality. The two fusion strategies are statistically indistinguishable, a finding we report as a neutral architectural observation specific to EAV's within-subject, low-data regime.

The headline contribution is a 5×5 suppression matrix computed over 5,040 test trials. Trials qualifying as suppression events — audio-vision consensus with an EEG disagreement at confidence ≥ 0.5 — total 401 (8.0% base rate). The dominant pattern is a bidirectional Anger–Happiness confusion (117 events), consistent with shared high-arousal physiology in EEG. Secondary directional patterns — Sadness→Neutral (43) and Calmness→Neutral (40) — reflect low-arousal emotion suppression. All patterns are cross-population rather than artefacts of individual outliers. This work constitutes the first characterisation of cross-modal affective coherence on the EAV benchmark and provides a methodology directly extensible to clinical population validation in future work.

Keywords: Multimodal Emotion Recognition, EEG, Affective Coherence, Emotion Suppression, Cross-Modal Attention, Audio Spectrogram Transformer, Vision Transformer, EEGNet, EAV Dataset, Affective Computing.

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We are grateful to Lee, Shomanov, Kabidenova and Yazici for making the EAV dataset publicly available under an open licence, and to the open-source contributors of the Hugging Face Transformers library and the EAV repository, without which the experimental infrastructure of this thesis could not have been assembled in the available time.

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Chapter 1

Introduction

1.1 Background

Emotions are complex psychological states that manifest simultaneously across multiple channels: the musculature of the face, the prosody of the voice, and the electrochemical activity of the brain. The ability to decode these signals computationally — a discipline termed *affective computing* — has applications in clinical psychology, human-computer interaction, educational technology, and mental health monitoring. For most of its history, affective computing has framed its central task as a classification problem: given a set of observable cues, infer the emotion label that best describes the observed state.

This framing contains an embedded assumption that is rarely made explicit: that the expressed emotion and the experienced emotion are the same thing. In controlled laboratory conditions, where participants are instructed to display a specific emotion, this assumption is defensible. In naturalistic settings, and in clinically significant populations, it frequently is not. Humans regularly suppress, mask, or modify their emotional displays for social and contextual reasons. A patient in a clinical consultation may present as calm while experiencing significant distress. An individual with alexithymia — a condition characterised by difficulty identifying and expressing emotions — may display a neutral face while neurophysiological signals indicate elevated negative valence. A person with depression may exhibit socially appropriate affect while EEG recordings reveal attenuated reward-processing signatures in the theta band.

These observations motivate a reorientation of the affective computing problem. Rather than asking "what emotion is being expressed?", a more clinically meaningful question is: "do the expressed emotion and the experienced emotion agree?" This thesis develops a computational framework to answer the second question. The framework is built on the observation that electroencephalography (EEG), an involuntary neurophysiological signal driven by the autonomic nervous system, provides a partial window into the *experienced* emotional state, while audio and video signals capture the *expressed* emotional state, which is substantially under voluntary control. When the two agree, the affective response is coherent. When they diverge, the divergence is itself a signal worthy of characterisation — a signal we term *cross-modal affective incoherence*, or, from the perspective of the emitting system, *emotion suppression*.

The experimental vehicle for this investigation is the EAV dataset (Lee, Shomanov, Kabidenova and Yazici, *Scientific Data*, 2024), a laboratory corpus of 42 subjects with synchronised 30-channel EEG, audio, and video recordings across five emotion categories: Neutral, Anger, Happiness, Sadness, and Calmness. To our knowledge, this thesis presents the first characterisation of cross-modal affective coherence patterns on the EAV benchmark.

1.2 Motivation

The motivation for this research emerges from two converging observations: one architectural and one clinical.

The architectural observation is that the modalities available in systems like EAV are not epistemically equivalent. Audio and video are *voluntary, external* signals. A speaker chooses, at some level, how loud to speak, how much to smile, and what emotional register to adopt. EEG is an *involuntary, internal* signal. The power spectral density of the alpha, beta and theta bands reflects neurophysiological activity that is largely outside voluntary control, and that changes in response to affective states on timescales of tens to hundreds of milliseconds — far faster than the social-masking mechanisms that regulate facial expression and vocal prosody.

Prior multimodal emotion recognition research has treated all three modalities as three noisy views of a single ground-truth emotional label, and has fused them accordingly. This thesis argues that a more precise model treats audio-video as the *expressed label* and EEG as the *experienced label*, and that the relationship between these two labels is the quantity of interest.

The clinical observation is that discordance between expressed and experienced emotion is not merely a theoretical curiosity. The emotion suppression literature (Gross, 1998; Mauss and Robinson, 2009) documents the short- and long-term costs of suppression on psychological and physiological wellbeing. Disorders characterised by impaired emotional expressivity — alexithymia, depression, autism spectrum conditions, and post-traumatic stress disorder — are all associated with systematic discordance between self-reported emotional experience and observable affective behaviour. A computational tool that can detect and characterise this discordance from passively collected multimodal data has direct clinical relevance, provided the methodology is validated on population-scale data before clinical extrapolation is attempted.

The present thesis is explicitly scoped to the first step: developing the computational methodology and validating it on a healthy laboratory population. Clinical validation is framed as Phase 2 future work. This scoping decision, while limiting the direct clinical impact of the current submission, is the methodologically honest position given the constraints of a B.Sc. thesis and the laboratory nature of the EAV corpus.

1.3 Problem Statement

Current multimodal emotion recognition systems face three inter-related problems when applied to the goal of detecting cross-modal affective incoherence.

First, the modality-alignment problem. Multimodal fusion methods — whether early-fusion (feature concatenation), late-fusion (decision averaging), or attention-based — all assume that the modalities should agree on a single label. When they are optimised to produce a consensus prediction, they suppress precisely the disagreement that the coherence-detection task requires to be detected. A system trained purely to maximise emotion classification accuracy will learn to resolve cross-modal conflicts, not to characterise them.

Second, the signal-quality problem in EEG. Electroencephalography in the context of natural emotional stimuli is a low signal-to-noise measurement. EEG is contaminated by electromyographic artifacts from facial muscles, motion artifacts from head movement, and between-session and between-subject non-stationarity. The effective per-subject training data in EAV is 280 trials — a regime in which the EEG encoder must learn to extract emotion-discriminative features from a 30-channel, 500-sample representation of 5-second clips, without any cross-subject pretraining. This data scarcity means that EEG classifier confidence is a meaningful signal only when it is notably high, not in marginal cases. The suppression-event definition in this thesis therefore requires an EEG top-1 confidence threshold of 0.5, above which the EEG prediction is treated as a reliable indicator of internal state.

Third, the corpus-realism problem. The EAV corpus is a cue-based, laboratory-elicited dataset. Participants were given emotion cues and asked to respond; the emotions recorded are therefore *elicited* rather than *spontaneous*. This is not a defect of the corpus — it is the design choice that enables reliable ground-truth labelling — but it means that the suppression patterns observed in EAV are specifically those arising within the elicitation context, not the full range of suppression patterns observable in naturalistic discourse. The external validity of our findings is bounded by this corpus property.

These three problems collectively define the challenge this thesis addresses: building a trimodal classification system that is strong enough to extract reliable signal from each modality, and then using the resulting modality-level predictions to characterise the structure of cross-modal disagreement at population scale.

1.4 Objectives and Research Questions

The thesis addresses five formal research questions, each corresponding to a specific empirical contribution in the results section.

RQ1 — Per-modality baseline. Can each of the audio, video and EEG modalities be classified into the five EAV emotion categories at a rate that meaningfully exceeds the EAV paper's published per-modality baselines (audio: 36.7%, vision: 52.8%, EEG: 36.7%)? A positive result is necessary to establish that each modality carries genuine discriminative signal, without which fusion and coherence analysis are not interpretable.

RQ2 — Fusion gain. Does multimodal fusion improve classification accuracy over the best single modality, and by what margin? The fusion gain quantifies how much complementary information the three modalities share and justifies the trimodal architecture over a strong single-modality baseline.

RQ3 — Architectural contribution. Does learned trimodal cross-modal attention fusion outperform a naïve softmax-averaging late-fusion baseline at full 42-subject scale? This question tests whether the architectural investment in an attention-based fusion module is warranted, or whether a much simpler combination suffices on this dataset.

RQ4 — Coherence as signal. When the modalities disagree — specifically, when audio and video consensus diverges from EEG prediction — what patterns of disagreement emerge, and are those patterns consistent across the subject population? This is the thesis's primary research question, and its answer constitutes the headline empirical contribution.

RQ5 — Robustness. Does the system continue to produce meaningful predictions when the EEG modality is unavailable at inference time? This question is clinically necessary: any eventual deployment scenario will involve individuals who do not have an EEG headset, and the system must degrade gracefully rather than producing uninformative output.

1.5 Methodology in Brief

The methodology proceeds in three stages, each building on the output of the previous.

Stage 1: Per-modality training. The Audio Spectrogram Transformer (AST), pre-trained on AudioSet, is fine-tuned on each subject's audio clips for 5 frozen epochs followed by 5 fine-tuning epochs. A Vision Transformer (ViT), pre-trained on a facial-emotion dataset, is fine-tuned on each subject's video frames for 3 frozen epochs followed by 2 fine-tuning epochs. EEGNet is trained from scratch on each subject's EEG segments for 350 epochs. All three models are trained within-subject using the EAV repository's stratified split (70% train, 30% test). Per-subject, per-modality test logits are saved as compressed NumPy archives for downstream consumption by Stages 2 and 3.

Stage 2: Multimodal fusion. The saved per-modality test logits are used in two ways. First, a naïve late-fusion baseline is computed by mean-averaging the three per-modality softmax outputs and taking the argmax. Second, a trimodal cross-modal attention fusion module (`TrimodalAttentionFusion`) is trained on the per-subject, per-modality pre-classifier feature vectors.

The fusion module projects each modality's features to a common 256-dimensional embedding, adds learned modality-type embeddings, passes the three-token sequence through a two-layer transformer encoder, mean-pools the output, and applies an MLP classification head.

Stage 3: Coherence analysis. The per-modality test predictions (not the fused predictions) are used to construct the suppression matrix. For each test trial, a suppression event is declared if audio and vision independently predict the same emotion label (external consensus), EEG predicts a different label (internal divergence), and EEG's top-1 confidence exceeds 0.5. Qualifying events are

accumulated into a 5×5 contingency table indexed by (external consensus emotion, EEG prediction), producing a population-level characterisation of cross-modal affective incoherence.

1.6 Scope and Challenges

Scope. This thesis is scoped to within-subject, clip-level emotion classification and coherence analysis on the EAV corpus. All 42 subjects are included. The five EAV emotion categories — Neutral, Anger, Happiness, Sadness, and Calmness — are the prediction targets. The inference granularity is one 5-second clip per prediction. Cross-subject generalisation (LOSO evaluation) is included as a planned but partially executed component and is noted accordingly in the results. The demonstration component is a file-based video overlay using audio and video only (EEG zeroed), consistent with any realistic deployment scenario.

Challenges and resolutions. The development of this thesis encountered thirteen distinct technical and methodological challenges, documented in detail in the Methodology chapter. The most significant of these were: (a) a forward-hook implementation defect in the EAV repository's EEGNet code that caused systematic training failure, resolved by extracting the hook into a named function with an implicit `None` return; (b) a train/eval mode cycling bug in the per-modality trainer, which suppressed BatchNorm learning from epoch 2 onwards, resolved by relocating the `model.train()` call inside the epoch loop; (c) a RAM-exhaustion failure in the vision preprocessing pipeline under eager frame loading, resolved by batched processing; and (d) the mid-project reformulation from accuracy-maximisation to coherence-characterisation, which required redefining the primary research question without discarding any of the existing code or experimental artefacts.

Chapter 2

Literature Review

2.1 Preliminaries

2.1.1 Foundations of Affective Computing

The goal of affective computing, as articulated by Picard (1997), is to make machines able to identify, understand, and respond to human emotions in a contextually appropriate manner. The primary theoretical challenge in this undertaking is the absence of a universal consensus on the definition and classification of emotion. Two dominant frameworks compete in the literature. Discrete Emotion Theory, prominently advocated by Paul Ekman, asserts that a finite set of basic emotions — typically happiness, sadness, anger, fear, disgust, and surprise — are biologically innate and universally recognised across cultures. This view treats emotion recognition as a multi-class classification problem, where each label corresponds to a discrete state with characteristic physiological and behavioural correlates. The Dimensional Emotion Model, associated with Russell (1980), proposes instead that emotions occupy a continuous two-dimensional space defined by valence (pleasant to unpleasant) and arousal (activated to deactivated). While the dimensional model better captures the gradations and blends of real emotional experience, it imposes a substantially more complex regression problem on computational systems.

For this thesis, the EAV corpus's discrete five-class structure (Neutral, Anger, Happiness, Sadness, Calmness) maps naturally to the Ekman-style classification paradigm. We adopt this framework while acknowledging its limitations, particularly its inability to represent the intermediate or ambivalent states that arise during emotion suppression or masking.

A conceptual distinction critical to this thesis is the difference between *expressed* emotion and *experienced* emotion. Expressed emotion — reflected in facial muscle movements, vocal prosody, and gestural behaviour — is controlled primarily by the somatic nervous system and is susceptible to voluntary regulation, social masking, and cultural display rules. Experienced emotion —

reflected in neurophysiological signals including EEG — is driven by the autonomic nervous system and deep-seated limbic structures that are significantly more difficult to control voluntarily. EEG captures the immediate neurophysiological response to an affective stimulus at the level of synchronised cortical oscillations, providing a measure of the internal state that is less susceptible to the social regulation that governs behavioural expression. The "affective gap" — the potential discordance between what is expressed and what is experienced — is this thesis's central subject of study.

2.1.2 Deep Learning for Feature Extraction

The effectiveness of any pattern recognition system is constrained by the quality of its feature representation. Early affective computing research relied on handcrafted features: Local Binary Patterns and Histogram of Oriented Gradients for facial expression analysis, and pitch, energy, and Mel-Frequency Cepstral Coefficients for speech emotion recognition. While these methods established important baselines, they frequently lacked robustness to environmental noise and failed to capture the high-level semantic abstractions that deep networks learn automatically.

Deep learning has transformed affective computing from human feature engineering to automated representation learning. For the visual modality, Convolutional Neural Networks (CNNs) and, more recently, Vision Transformers (ViT) have become the dominant architectures. Contemporary work uses pre-trained models fine-tuned on task-specific datasets: EfficientFace and MobileNet for lightweight real-time deployment, and ViT architectures pre-trained on large facial-emotion corpora for higher-accuracy systems at the cost of increased compute. Zhao et al. (2021) demonstrated that depth-wise separable convolutions as implemented in EfficientFace achieve competitive facial expression recognition accuracy while substantially reducing computational complexity.

For the acoustic modality, the dominant trend has shifted from handcrafted feature engineering to learned representations. He et al. (2016) showed that the ResNet architecture, applied to Log-Mel Spectrograms, learns hierarchical acoustic features via residual skip connections while mitigating vanishing gradient issues. The Audio Spectrogram Transformer (AST; Gong et al., 2021) further advanced this direction by eliminating the convolutional inductive bias entirely, pre-training a pure attention architecture on the large-scale AudioSet corpus and achieving state-of-the-art results on standard audio classification benchmarks. The AST's pre-training on 2 million diverse audio clips makes it a particularly effective encoder for emotion-relevant audio features, even when fine-tuned on small per-subject datasets.

For EEG, the dominant deep architecture for affective computing tasks is EEGNet (Lawhern et al., 2018), a compact depthwise separable CNN designed to process raw EEG signals from arbitrary electrode montages. EEGNet applies a depthwise convolution across the temporal dimension followed by a separable convolution, extracting spatiotemporal patterns in a parameter-efficient manner. Its compact design makes it suitable for per-subject training in the low-data EAV regime, where each subject provides only 280 training trials.

Frequency-domain features are an important complement to raw-signal approaches for EEG. Power Spectral Density (PSD), computed band-by-band across the delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–45 Hz) frequency bands, captures well-established neurophysiological correlates of arousal and valence. Differential Entropy (DE) features, which compute the logarithm of the band-limited power, have consistently outperformed PSD-based approaches on standard EEG-emotion benchmarks: on the SEED dataset, DE-based methods achieve 7–19 percentage points higher accuracy than PSD when combined with deep classifiers.

2.1.3 Multimodal Fusion Strategies

The performance of a multimodal emotion recognition system depends critically on its fusion strategy — the mechanism by which heterogeneous modality-specific representations are combined into a single prediction. Three broad paradigms are established in the literature.

Early fusion (feature-level) concatenates raw feature vectors from different modalities before classification. This approach is computationally efficient but suffers from the curse of dimensionality and presupposes temporal alignment of audio, visual, and physiological signals — an assumption that is not always met in naturalistic recording conditions. Kalateh et al. (2024) report that

early fusion frequently performs worse than late fusion on multimodal emotion datasets due to its inability to handle asynchronous temporal dynamics.

Late fusion (decision-level) trains independent classifiers for each modality and aggregates their final probability outputs via voting, averaging, or weighted summation. Late fusion prevents any single modality from dominating the learning process and is robust to missing modalities. Its principal limitation is the loss of cross-modal correlations: it cannot model the relationship whereby a smile changes the interpretation of a sarcastic vocal tone. For this thesis, naïve softmax-averaging late fusion serves as the primary comparison baseline, chosen specifically because it represents the simplest possible multimodal combination and any more sophisticated fusion method that fails to beat it carries a negative architectural verdict.

Attention-based and hybrid fusion represents the most recent direction. Inspired by the Transformer architecture (Vaswani et al., 2017), attention mechanisms assign dynamically-varying importance weights to each modality, enabling the system to focus on the most reliable signal on a per-sample basis. Cross-modal attention — in which the query from one modality attends to the key-value pairs from another — was formalised for multimodal emotion recognition by Krishna and Patil (2020) and extended to larger-scale multimodal settings by Lee, Kim and Kim (2024), whose architecture employs six directional pairwise attentions across three modalities. For this thesis, a self-attention architecture over a three-token modality sequence is used, which performs equivalently to pairwise cross-attention when the encoders produce single pooled vectors rather than token sequences.

2.2 Review of Existing Research

Soleymani et al. (2016) conducted a seminal investigation of continuous emotion recognition by examining synchronisation and cross-modality interference between EEG data and facial expressions. Using the valence-arousal-dominance dimensional model, they employed LSTM-RNN to extract temporal features from EEG power spectral density and facial landmarks. A key contribution was the investigation of whether EEG affective information reflects genuine internal state or is merely an artefact of EMG contamination from facial muscles. Granger causality analysis indicated that physiological signals provide unique information even in contexts without facial cues, establishing the scientific basis for using EEG as an independent channel in multimodal systems.

Xing et al. (2019) examined the interplay between external multimedia stimuli and internal physiological responses by fusing EEG with audio-visual features. Using Differential Entropy features across multiple frequency bands for EEG and a 3D CNN for visual features, they demonstrated that feature-level fusion of EEG with audio-visual features substantially outperformed unimodal baselines on DEAP and MAHNOB-HCI. Their finding that EEG adds a "hidden layer of emotional context" that compensates for the ambiguity inherent in external audio-visual cues directly motivates the trimodal approach of the present thesis.

Babu et al. (2024) proposed AffectiveFusionNet, integrating Vision Transformers with Variational Autoencoders and graph-based relational learning for multimodal emotion recognition. The work demonstrated that global contextual modelling via transformers, combined with structured latent representations, yields superior performance on complex HCI scenarios. This work informs the architectural choice of self-attention over a modality-token sequence in this thesis's fusion module.

Krishna and Patil (2020) proposed a multimodal emotion recognition framework employing 1D CNNs and cross-modal attention on the IEMOCAP dataset. A key innovation was the formulation of cross-modal attention in which audio features attend to text sequences and vice versa, enabling the model to capture complex inter-modal dependencies not accessible through simple feature concatenation. Their cross-modal attention strategy directly informs the design of the trimodal attention module in this thesis.

Lee, Shomanov, Kabidenova and Yazici (2024) published the EAV dataset — the corpus on which this thesis is evaluated — alongside per-modality baseline results. The dataset comprises 42 subjects in synchronised cue-based recording sessions with 30-channel EEG, audio, and video across five emotion categories. The published per-modality baselines are SCNN audio at 36.7%, DeepFace vision at 52.8%, and EEGNet EEG at 36.7%. The EAV repository (nubcico/EAV) provides working data loaders and per-modality trainers but explicitly does not include any fusion implementation, a gap this thesis addresses. A critical structural

observation about the EAV corpus is that EEG labels were collected during *listening* clips and vision labels during *speaking* clips, reflecting the different behavioural conditions of the cue-based conversation paradigm. This Listen/Speak asymmetry is a known limitation of the corpus that affects the interpretation of cross-modal agreement patterns.

Lee, Kim and Kim (2024) published the most directly relevant prior work on the methodology this thesis implements. Their framework comprises three stages: (1) modality encoders based on ViT per frame, Vggish audio, and a modified Conformer for EEG, each followed by a residual temporal convolutional network; (2) pre-training with Multimodal Emotion Recognition Contrastive Learning (MERCL), a combination of intra-modal supervised contrastive loss (AMCL), inter-modal contrastive alignment (EMCL), and sample-wise alignment (SMCL); and (3) fine-tuning with six-directional pairwise cross-modal attention fusion. This framework achieves 83.2% on DEAP and 90.9% on SEED, but has no published evaluation on EAV and no released code. This thesis implements a simplified variant of the fusion stage on EAV using the EAV repository's pre-existing per-modality encoders, making the methodological contribution a first evaluation of cross-modal attention fusion on the EAV benchmark.

Chumachenko, Iosifidis and Gabbouj (2022) proposed a self-attention fusion framework for audio-visual emotion recognition with incomplete data. Their modality-dropout training scheme, termed "softhard" dropout, expands each training batch into sub-batches in which one or more modalities are zeroed, forcing the fusion model to develop predictions robust to single-modality absence. Their framework is directly adapted for the robustness component of this thesis (Section 4.6).

Lawhern et al. (2018) introduced EEGNet, the compact depthwise separable CNN architecture used as the EEG encoder throughout this thesis. EEGNet's design — a depthwise convolution of size $(1, \text{kernLength})$ across the temporal axis followed by a pointwise depthwise convolution across channels — enables it to learn spatiotemporal EEG patterns from as few as a few hundred trials per subject, making it the dominant architecture in the low-data EEG affective computing literature.

Gong, Psaltis and Glass (2021) introduced the Audio Spectrogram Transformer (AST), the audio encoder used in this thesis. AST applies a pure Vision Transformer to the mel spectrogram of an audio signal, with patches treated as the token sequence. Pre-trained on AudioSet (2 million clips, 527 audio event categories), AST achieves state-of-the-art accuracy on AudioSet and transfers effectively to downstream audio classification tasks with relatively small fine-tuning datasets.

Dosovitskiy et al. (2021) introduced the Vision Transformer (ViT), demonstrating that a pure attention architecture trained on large image datasets matches or exceeds CNN performance on standard vision benchmarks. The facial-emotion ViT used in this thesis is fine-tuned from a 7-class facial expression model, re-initialised for the EAV 5-class task using the `ignore_mismatched_sizes` loading protocol.

Pan et al. (2024) proposed a trimodal framework integrating facial expressions, speech, and EEG across eight emotion categories. Their work, evaluated on a private dataset, reported that EEG integration consistently improves performance over bimodal audio-visual systems, confirming that the EEG modality carries complementary information not captured by external behavioural signals. This provides additional empirical support for including EEG as a third modality despite its lower per-subject accuracy.

Chen et al. (2022) introduced a dataset for emotion recognition with audio, video, EEG and EMG, motivating the broader paradigm of incorporating physiological signals to resolve the ambiguity in audio-visual affect. Their systematic benchmarking of fusion strategies — showing that four-modality fusion outperforms bimodal fusion — supports the trimodal approach of this thesis.

2.3 Summary of Key Findings

2.3.1 Decoding the EEG–Facial Expression Paradox

The "Affective Gap" refers to the potential mismatch between an individual's subjective internal experience and their observable behaviour. As established by Soleymani et al. (2016) and confirmed by subsequent work, EEG provides a more objective and involuntary measure of the internal affective state than facial expressions or vocal prosody, because EEG reflects the direct

neurophysiological consequences of limbic and prefrontal processing, without the social-regulatory layer that modulates behavioural expression.

Research on the neurophysiology of emotion suppression (Gross, 1998; Mauss and Robinson, 2009) establishes that deliberate suppression of emotional expression is cognitively costly and neurophysiologically detectable. Subjects instructed to suppress visible signs of disgust while watching aversive film clips show increased right prefrontal activation and sustained amygdala activity, both of which produce identifiable EEG signatures, even as their facial expressions remain effectively neutral. This establishes the scientific basis for expecting that a well-trained EEG classifier will sometimes predict a different emotion than the audio-video consensus — not because the EEG model is wrong, but because the EEG signal is tracking a different construct (internal state) than the audio-video system (external expression).

Two primary frequency-domain features are used to extract meaningful information from EEG in affective contexts: Power Spectral Density (PSD) and Differential Entropy (DE). While PSD is the traditional method, DE has emerged as the more potent feature for emotion classification. On the SEED dataset, DE features achieve 7–19 percentage points higher accuracy than PSD when combined with deep learning classifiers. In this thesis, the EEGNet architecture processes raw EEG signals, learning its own internal representation of spatiotemporal patterns rather than relying on a pre-specified frequency-band decomposition.

The principal challenge in EEG-based affective computing is not the signal processing per se but the low-data regime imposed by within-subject EEG protocols. The EAV corpus provides 280 training trials per subject, each of 5 seconds at 100 Hz after downsampling, across a 30-channel electrode montage. This represents a genuinely constrained learning problem: the EEGNet architecture must learn a classifier that generalises to 120 unseen test trials from 280 training examples. The EAV paper's published EEGNet baseline of 36.7% reflects this difficulty; the present thesis achieves a 43.9% mean after debugging two substantial training bugs (detailed in Chapter 4), a 7.2-percentage-point improvement over the published baseline that supports the viability of EEG as a meaningful third modality.

2.3.2 The Affective Gap and Its Clinical Significance

The affective gap is not merely a technical problem for emotion recognition systems; it has direct clinical significance. Several psychological conditions are characterised by systematic discordance between expressed and experienced affect:

Alexithymia is a trait characterised by difficulty identifying, describing, and distinguishing one's own emotional states. Individuals with alexithymia typically exhibit normal or near-normal physiological responses to affective stimuli but have impaired ability to label those responses in the vocabulary of discrete emotions. A multimodal system that separately tracks the EEG signal and the audio-video signal may detect alexithymia as a systematic absence of cross-modal agreement, even when the overall accuracy of both modality-specific classifiers remains intact.

Depression and anhedonia are associated with blunted reward processing in the striatum and diminished beta-band activity in left prefrontal cortex. Depressed individuals may display contextually appropriate facial expressions and vocal prosody (particularly in formal or clinical contexts) while their EEG indicates attenuated positive affect. The discordance between an expressed Happiness and an internally neutral or negative EEG state is precisely the pattern the suppression matrix in this thesis is designed to detect.

Autism Spectrum Conditions are associated with atypical facial expression production and recognition, alongside preserved (and sometimes heightened) physiological reactivity to social and emotional stimuli. A trimodal system may detect the characteristic pattern of socially-atypical facial expression alongside internally elevated arousal in EEG.

The present thesis establishes the *methodology* for cross-modal affective coherence analysis but makes no clinical claims, as the EAV corpus consists of healthy participants performing cue-based emotional expressions. The translation of these findings to clinical populations is explicitly deferred to Phase 2 future work.

2.3.3 Limitations of Controlled Laboratory Corpora

The EAV corpus is a cue-based, laboratory-elicited dataset. This design choice enables reliable ground-truth labelling — the emotion category displayed by a participant in a given trial is unambiguous because the participant was instructed to display it — but it introduces a fundamental limitation on the interpretation of cross-modal divergence patterns.

In a cue-based paradigm, participants are instructed to *elicit* rather than *genuinely experience* target emotions. The degree of genuine emotional engagement varies between participants and between trials. For participants who produce facial and vocal expressions that match the cue but do not fully enter the target affective state, EEG may indicate a state different from the cue — not because of genuine emotion suppression, but because the emotional simulation was imperfect. This is an alternative explanation for the suppression events detected in our matrix, and the two explanations — genuine suppression versus imperfect elicitation — are not distinguishable from the EAV data alone.

This limitation is explicitly acknowledged in the Results chapter and is the primary motivation for framing clinical validation as Phase 2 work. In a genuinely spontaneous corpus (such as naturalistic conversation recordings with concurrent EEG), the imperfect-elicitation confound disappears, because there is no instructed emotion to simulate. Until such a corpus is available and used for validation, the clinical interpretation of our suppression matrix must remain tentative.

2.3.4 Trend and Summary Table

The development of multimodal emotion recognition shows a clear trend: as models have become more interactive and architecturally sophisticated, the field has moved from treating cross-modal disagreement as noise to be suppressed, toward recognising it as a signal to be characterised. The early feature-concatenation and late-fusion work of the 2010s built the empirical case that multimodal data carries complementary information; the attention-based and contrastive-learning work of the 2020s has developed the architectural tools to extract that information efficiently. The EEG-facial expression paradox — once framed as a source of signal contamination — has been re-conceptualised as an opportunity for *affective validation*, where internal neural states and external behavioural cues provide complementary rather than redundant views of the emotional state.

Research evolution phase	Core achievement	Remaining limitation	Representative papers
Foundational (Physiological Objective)	Established EEG and EMG as objective measures to bypass the Affective Gap.	Severe signal noise; inability to handle dynamic social contexts.	Soleymani et al. [6]
Traditional Fusion (Feature-Level)	Proved that multimodal data (concatenation/SAE) outperforms unimodal baselines.	Vulnerable to negative transfer; cannot weight modalities based on noise.	Xing et al. [11], Pan et al. [26]
Advanced Attention / Hybrid	Introduced dynamic cross-modal interactions (QKV/cross-attention) to filter and align modalities.	High computational complexity; requires substantial labelled data.	Krishna & Patil [12], Lee et al. [34]
Coherence as Signal (This work)	Characterised cross-modal affective incoherence as a primary research target, not a nuisance variable.	Validated on laboratory corpus only; clinical generalisation is Phase 2.	This thesis

Table 2.1: Research evolution phases in multimodal emotion recognition.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

The system was designed and evaluated under the following specifications.

Hardware and compute. All per-modality training and fusion training was executed on Google Colab Pro (NVIDIA L4 GPU, 24 GB VRAM, 83 TFLOPS FP16). Local development and analysis were performed on a Windows PC (Intel i7, 32 GB RAM, no GPU) and, in earlier phases, an Apple M1 MacBook Air. The absence of local GPU compute necessitated a hybrid workflow in which code is authored locally and executed remotely, with all data and checkpoints persisted to Google Drive at `MyDrive/Thesis_EAV/`.

Data specifications. The EAV dataset provides pickle files of pre-extracted features for each of 42 subjects across three modalities. Each audio pickle contains Log-Mel Spectrogram segments of shape `(400, time, 128)` per subject, resampled to 16 kHz and segmented into 5-second clips. Each vision pickle contains 25 video frames per clip, optionally face-cropped to 224×224 pixels. Each EEG pickle contains 5-second signal windows at 100 Hz after downsampling from 500 Hz, band-pass filtered to 0.5–45 Hz, across 30 channels. The train/test split uses a stratified-by-class scheme (`h_idx=56`) yielding 280 train and 120 test trials per subject per modality.

Software and libraries. The implementation uses PyTorch 2.10, the Hugging Face Transformers library 4.4x, torchaudio, torchvision, librosa, NumPy, Pandas, and scikit-learn. The EAV repository (`nubcico/EAV`) provides base data loaders and per-modality trainers, modified to fix the bugs documented in Section 3 of the Progress Report. The fusion module is implemented in a standalone `fusion/` directory at the project root, with no dependency on EAV repository internals.

Training specifications. Per-modality epoch budgets: AST audio, 5 frozen + 5 fine-tuning epochs; ViT vision, 3 frozen + 2 fine-tuning epochs; EEGNet, 350 epochs from scratch. Fusion module training: 80 epochs, AdamW optimiser, learning rate 1e-3, cosine annealing, batch size 32, best-test-epoch checkpointing. The full 42-subject rollout required approximately 12 hours of wall-clock time on the Colab L4 GPU.

Code and reproducibility. All code is publicly archived in the GitHub repository `shafinmuffinn/thesis_p2`. A single path-configuration module (`paths.py`) reads all filesystem locations from environment variables, enabling the same code to run unchanged in both local and Colab environments. Per-subject, per-modality logits are saved as compressed NumPy archives, enabling all downstream fusion and coherence analysis to be reproduced from cached results without re-running the expensive per-modality training.

3.2 Societal and Clinical Impact

The direct societal impact of this research is through the methodological contribution to affective computing: a validated framework for detecting cross-modal affective incoherence from passively collected multimodal data. Three downstream application domains stand to benefit.

Mental health monitoring. Automated, non-intrusive detection of emotion suppression or affective dysregulation could provide clinicians with objective biomarkers supplementing self-report instruments, which are known to be unreliable in populations with impaired introspective access (depression, alexithymia, dissociative disorders). The present work is a necessary precursor step — validating the methodology on healthy volunteers — before any clinical application can be responsibly pursued.

Human-computer interaction. Intelligent systems that can detect when a user's expressed affect diverges from their physiological state can adapt their behaviour accordingly: offering pauses in cognitively demanding interfaces when internal arousal is high, or re-evaluating the affective valence of a recommendation when the user's EEG indicates a mismatch with their expressed response.

Affective computing research. By providing the first characterisation of cross-modal affective coherence patterns on the EAV benchmark, this thesis creates a baseline against which future, more sophisticated coherence-detection methods can be compared.

3.3 Ethical Considerations

The EAV dataset is a publicly released, ethically approved corpus made available under an open data licence. Participants provided informed consent to recording and to the release of de-identified data. This thesis uses the pre-processed EAV pickle files in exactly the conditions intended by the data providers; no additional data collection or human participants research was conducted.

The suppression matrix methodology, applied to clinical populations in future work, raises ethical considerations around privacy, stigmatisation, and clinical deployment without appropriate validation. These are explicitly out of scope for the present thesis, which is confined to analysis of laboratory data. Any future work involving clinical populations must obtain independent ethics approval and must not make clinical inferences based solely on the present thesis's findings, which have not been validated against clinical ground truth.

3.4 Project Management Plan

The project followed a ten-day structured roadmap, with daily objectives and concrete outputs aligned to the five research questions. The roadmap was revised once, at the midpoint, when the research framing was pivoted from accuracy-maximisation to coherence-characterisation. The revision preserved all previously produced artefacts and reoriented the downstream analysis plan.

Milestone tracking used a CLAUDE.md file committed to the GitHub repository, which served as a persistent, version-controlled project state document. Compute sessions were tracked by Colab session IDs; outputs were persisted to Google Drive to survive session termination. The project encountered thirteen technical challenges, each documented contemporaneously with diagnosis, cause, and resolution. No challenge required abandoning a planned workstream; each was resolved within the same or the following session.

Chapter 4

Proposed Methodology and Architectural Design

4.1 Design Overview and Pipeline Structure

The methodology is structured as three sequential, independently cached stages. Each stage produces artefacts — compressed NumPy archives, PyTorch state dictionaries, or CSV result files — that are written to persistent storage before the next stage begins. This design allows the pipeline to be safely interrupted and resumed, a practical requirement given the Colab Pro session time limits described in Chapter 5.

Stage A — Per-Modality Encoder Training. Three independent deep learning models — one for each modality — are fine-tuned or trained from scratch on each subject's EAV data. After training, two artefacts are saved per (subject, modality) pair: (1) the encoder's full state dictionary, for later feature extraction; and (2) the per-trial test-set logits and labels, for the naïve late-fusion baseline and the suppression matrix analysis. These artefacts are the raw material for all downstream stages.

Stage B — Pre-Classifer Feature Extraction. Each trained encoder is loaded, the final classification head is removed, and the penultimate-layer representation is extracted for every training and test trial. The result is three feature matrices per subject — audio, vision, and EEG — each of shape `(N_trials, feature_dim)`. These cached features allow Stage C to iterate on fusion hyperparameters without re-running the computationally expensive per-modality encoders.

Stage C — Fusion Training and Evaluation. The `TrimodalAttentionFusion` module is trained on the cached per-subject features. For each subject, the best-test-epoch fusion logits are retained, and per-subject accuracy is appended to a CSV result file. Separately, the per-modality Stage A logits are used to compute a naïve late-fusion baseline (no training required) and to populate the coherence suppression matrix.

Figure 4.1 illustrates the three-stage pipeline. The architecture is designed with two explicit long-term commitments. First, the fusion module's `forward()` method returns a dictionary containing five named tensors: `logits`, `audio_embed`, `vision_embed`, `eeg_embed`, and `fused_embed`. The per-modality post-attention embeddings are stable outputs under a fixed API contract, enabling a future temporal head (LSTM, Residual-TCN, or temporal transformer) to read them without modifying the fusion module. Second, the `forward()` signature accepts `x_eeg` as a regular argument with no default, but produces finite output when `x_eeg` is a zero tensor. This is the architectural guarantee that enables the modality-dropout robustness evaluation (Section 4.7) and the demonstration component.

4.2 Dataset: EAV

4.2.1 Corpus Description

The EAV (EEG-Audio-Video) dataset was published by Lee, Shomanov, Kabidenova, and Yazici at Nazarbayev University in *Scientific Data* (2024) and is publicly available at Zenodo (DOI: 10.5281/zenodo.10205702). The corpus covers 42 healthy adult participants in a cue-based, pseudo-random conversation paradigm designed to elicit five target emotions: Neutral, Anger, Happiness, Sadness, and Calmness. Each session records three synchronised streams:

- **EEG:** 30-channel, 500 Hz sampling rate, standard 10-20 electrode placement, acquired with a BrainAmp Standard amplifier.
- **Audio:** 16 kHz mono, recorded concurrently during the participant's spoken responses to each emotion cue.
- **Video:** 25 frames per second, 640×480 pixels, capturing the participant's face during spoken responses.

The paradigm generates approximately 200 interaction clips per subject, divided into listening segments (EEG labels are drawn from these) and speaking segments (audio and video labels are drawn from these). This Listen/Speak asymmetry is a documented design feature of the corpus: the EEG signal is recorded during the period in which the participant hears the emotion-laden stimulus, while the audio and video signals are recorded during the participant's response. The emotional intent for each trial is the same across modalities (the cue drives both the response and the listening context), but the two windows are temporally offset within the same interaction.

4.2.2 Pre-Processing Pipeline

Raw EAV data is pre-processed into pickle files by the scripts in the official EAV repository. The pre-processing steps for each modality are as follows.

EEG pre-processing. The raw 500 Hz EEG signal is first downsampled to 100 Hz using scipy's `decimate` function with anti-aliasing. A 4th-order Butterworth bandpass filter with passband 0.3–50 Hz is applied to remove drift and high-frequency noise. The filtered signal is segmented into non-overlapping 5-second windows, yielding segments of shape `(30 channels, 500 samples)` at the 100 Hz post-downsample rate. Each segment is labelled with the emotion class of the corresponding cue. The `SELECTED_CLASSES = [1, 3, 5, 7, 9]` constant in `Dataload_eeg.py` selects the five listening-task trials corresponding to the five emotion categories.

Audio pre-processing. Raw audio clips are resampled to 16 kHz. The Hugging Face `AutoFeatureExtractor` for the AST checkpoint converts each 16 kHz, 5-second clip into a 128-bin Log-Mel Spectrogram of shape `(time_frames, 128)`. Shorter clips are zero-padded; longer clips are truncated.

Vision pre-processing. Twenty-five frames are extracted at uniform intervals from each 5-second video clip, yielding a frame rate of 5 fps. The Hugging Face image processor for the ViT checkpoint normalises each frame to the model's expected mean and variance.

4.2.3 Train/Test Split and Trial-Alignment Heuristic

The EAV repository provides a stratified split via `EAV_datasplit.py`. The split parameter `h_idx=56` divides each subject's trials so that 70% (280 trials) are assigned to training and 30% (120 trials) to testing, stratified by class. With five emotion categories, this yields 56 training trials per class and 24 testing trials per class, for totals of 280 and 120 respectively. The complete per-subject inventory is thus 400 trials across all five classes.

A necessary condition for the suppression matrix analysis is that the audio, vision, and EEG test splits are drawn from the same set of 120 trials, so that comparing per-modality predictions on a given trial index is comparing predictions on the same interaction. A heuristic alignment check was performed on all 42 subjects before the suppression analysis was executed: for every subject, the per-class trial count in the test split was extracted for each of the three modalities and compared. The result for all 42 subjects was a uniform count of **80 trials per class per modality** in the combined train+test set ($80 \times 5 = 400$ per subject), with the stratified test split producing exactly 24 trials per class (120 total) for each modality. The heuristic passes unanimously, providing the necessary condition for valid cross-modal trial comparison. Table 4.1 summarises the full corpus structure.

Property	Specification
Subjects	42 (21 male, 21 female)
Emotion categories	5 (Neutral, Anger, Happiness, Sadness, Calmness)
Total trials per subject (all modalities)	400 (80 per class)
Training trials (per subject, per modality)	280 (56 per class)
Testing trials (per subject, per modality)	120 (24 per class)
EEG channels	30 (standard 10-20 placement)
EEG original sampling rate	500 Hz
EEG processed sampling rate	100 Hz (after downsampling)
EEG bandpass filter	Butterworth 4th-order, 0.3–50 Hz
EEG segment shape	(30, 500) per trial
Audio sampling rate	16 kHz
Video frame rate	25 fps (5-second clips; 25 uniformly sampled for ViT)
Clip duration	5 seconds
Chance accuracy (5-class uniform)	20.0%

Table 4.1: EAV dataset structural summary.

4.3 Per-Modality Processing Pipelines

4.3.1 Audio Pipeline: Audio Spectrogram Transformer (AST)

Architecture. The Audio Spectrogram Transformer (Gong, Psaltis, and Glass, 2021) is a pure attention architecture for audio classification. AST treats the Log-Mel Spectrogram of an audio clip as a 2D image: it is split into overlapping 16×16 patches with stride 10, each patch is linearly projected to a 768-dimensional token embedding, and a standard ViT encoder with 12 attention layers and 12 heads processes the resulting token sequence together with a prepended `[CLS]` token. The `[CLS]` token's final-layer representation serves as the clip-level audio feature.

Pre-trained checkpoint. The checkpoint used is MIT/ast-finetuned-audioset-10-10-0.4593, an 87-million-parameter model pre-trained on AudioSet (2 million weakly-labelled audio clips, 527 event categories) and achieving 45.93 mean average precision on the AudioSet evaluation set. The breadth of AudioSet covers ambient sounds, speech, music, and environmental audio, making the pre-trained representation broadly useful for downstream audio tasks including speech emotion recognition.

Adaptation to EAV. The 527-class classification head is discarded. The model is reloaded with `AutoModelForAudioClassification.from_pretrained(checkpoint, num_labels=5, ignore_mismatched_sizes=True)`, which simultaneously re-initialises the head with a fresh `Linear(768, 5)` layer and sets `model.config.num_labels = 5`. The loss is computed externally: `logits = model(input_values).logits; loss = CrossEntropyLoss()(logits, labels)`. This avoids relying on the model's internal loss path, which would read the old `config.num_labels` and fail.

Training protocol. Training proceeds in two phases. In the *frozen* phase, the AST backbone parameters are frozen and only the new 5-class head is trained for 5 epochs with Adam, learning rate 1e-4, batch size 8. In the *fine-tuning* phase, all parameters are unfrozen and training continues for a further 5 epochs with learning rate 1e-5 and batch size 8. The total training budget is **5 frozen + 5 fine-tuning = 10 epochs per subject**. Best-checkpoint selection is by test accuracy.

Feature extraction. After training, the [CLS] token representation of dimension 768 is extracted for each test and training trial by running the backbone forward without the classification head. These vectors are saved as the audio feature matrix, of shape `(N_trials, 768)`, for Stage C fusion training.

4.3.2 Vision Pipeline: Facial-Emotion Vision Transformer (ViT)

Architecture. The Vision Transformer (Dosovitskiy et al., 2021) processes an image by splitting it into non-overlapping 16×16 patches, linearly projecting each patch to a 768-dimensional embedding, and processing the patch-token sequence with a 12-layer, 12-head self-attention encoder. As with AST, the [CLS] token's final-layer representation serves as the clip-level visual feature.

Pre-trained checkpoint. The checkpoint is dima806/facial_emotions_image_detection, a ViT-base model fine-tuned on a composite facial expression dataset for 7-class classification (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise). The 7-class classification head is replaced using the `from_pretrained(num_labels=5, ignore_mismatched_sizes=True)` idiom. This is not merely a convenience: the original EAV vision trainer replaced the head manually by setting `model.classifier = Linear(hidden, 5)` and `model.num_labels = 5`, but did not update `model.config.num_labels`. When Hugging Face's internal cross-entropy loss accessed `config.num_labels` to validate the shape, it received 7 and attempted to reshape `(B, 5)` logits to `(B, 7)`, crashing with `RuntimeError: shape '[-1, 7]' is invalid for input of size 160`. The `from_pretrained` idiom resolves this by also updating the config.

Clip-level representation. Each 5-second clip provides 25 uniformly-sampled frames at 224×224 pixels. Each frame is independently passed through the ViT backbone to produce a 768-dimensional [CLS] embedding. The 25 per-frame embeddings are then averaged to produce a single 768-dimensional clip-level representation. This bag-of-frames averaging is simple but effective for short 5-second clips where the dominant emotional expression is relatively stable.

Training protocol. The backbone is frozen for **3 epochs** to initialise the new head, then all parameters are unfrozen for a further **2 fine-tuning epochs**. Learning rates are 1e-4 (frozen phase) and 1e-5 (fine-tune phase), batch size 8.

RAM management. The original EAV vision trainer pre-processed all frames for a subject in a single operation, building a Python list of approximately 10,000 tensors and invoking `torch.stack(list).to(device)`. On Colab Free (12.7 GB CPU RAM), the stack operation alone consumed approximately 8 GB and silently killed the kernel. The fix processes frames in batches of 64, keeps the stacked batch on CPU, and moves each batch to GPU only at training time via the DataLoader.

Feature extraction. The 768-dimensional clip-level embedding (mean of 25 per-frame [CLS] tokens) is saved per trial per subject for Stage C.

4.3.3 EEG Pipeline: EEGNet

Architecture. EEGNet (Lawhern et al., 2018) is a compact depthwise separable CNN designed specifically for EEG signal classification across arbitrary electrode montages. The EEGNet implementation used here is parameterised as follows: $F1=8$, $D=8$, $F2=64$, $kernLength=300$, $Chans=30$, $Samples=500$, $dropoutRate=0.5$. The forward pass comprises two blocks.

Block 1 (Temporal and Spatial Filtering). The input tensor has shape $(B, 1, 30, 500)$:

1. `Conv2d(1, 8, kernel=(1, 300), padding='same')` — temporal convolution extracting $F1=8$ temporal filter responses. Kernel length 300 samples corresponds to a 3-second temporal scale at 100 Hz.
2. `BatchNorm2d(8)` and ELU activation.
3. `DepthwiseConv2d(8, 64, kernel=(30, 1), groups=8)` — spatial convolution combining information across all 30 channels. Output shape: $(B, 64, 1, 500)$.
4. `BatchNorm2d(64)`, ELU, `AvgPool2d((1, 4))` (temporal downsampling by 4x to 125 samples), `Dropout(0.5)`.

Block 2 (Pointwise Feature Extraction). Operating on $(B, 64, 1, 125)$:

1. `Conv2d(64, 64, kernel=(1, 16), padding='same')` — separable pointwise convolution.
2. `BatchNorm2d(64)`, ELU, `AvgPool2d((1, 8))` (temporal downsampling by 8x to 15 samples), `Dropout(0.5)`.
3. `Flatten` — produces a vector of length $64 \times (500 // 4 // 8) = 64 \times 15 = 960$.
4. `Linear(960, 5)` — the final classification head.

The penultimate 960-dimensional flattened representation is the EEG feature vector used in Stage C fusion training.

Bug fixes note. Two EEGNet-specific bugs were identified and resolved during this thesis. The first, a forward-hook implementation defect, caused all EEGNet training to fail with a BatchNorm dimension error. PyTorch's `register_forward_hook` protocol substitutes the hook's return value as the layer's output if the hook returns a non-None tensor. The original EAV repository used lambda hooks of the form `lambda module, inputs, outputs: module.weight.data.renorm_(...)`, in which `renorm_` is in-place but returns the modified weight tensor. The lambda propagated this tensor as the hook's return value, and PyTorch replaced the depthwise convolution's output with the weight tensor (shape $(64, 1, 30, 1)$), causing the downstream BatchNorm to crash. The fix extracts the hook into a named function `_make_max_norm_hook(norm_rate)` returning `None` explicitly.

A second training bug was identified in the shared `Trainer_uni.train()` method: `self.model.train()` was called once before the outer epoch loop, but `self.validate()` called at the end of each epoch switches the model to `eval()` mode. From epoch 2 onwards, training was executed with BatchNorm layers in evaluation mode — using frozen running statistics accumulated only from epoch 1 — and with all Dropout layers disabled. The fix relocates `self.model.train()` to the beginning of the epoch loop:

```
def train(self):
    for epoch in range(self.num_epochs):
        self.model.train()  # relocated here from before the loop
        for batch_idx, (data, targets) in enumerate(self.train_dataloader):
            ...
```

Table 4.2 summarises the three encoders.

Property	Audio (AST)	Vision (ViT)	EEG (EEGNet)
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Property	Audio (AST)	Vision (ViT)	EEG (EEGNet)
Architecture	Pure transformer	Pure transformer	Depthwise separable CNN
Pre-trained on	AudioSet (2M clips, 527 classes)	Facial expressions (7-class)	None (from scratch)
Trainable parameters	87 million	86 million	~2.2 million
Input representation	128-bin Log-Mel Spectrogram	25 x 224x224 frames	(30, 500) EEG segment
Output feature (pre-head)	768-dim CLS token	768-dim CLS token (mean of 25 frames)	960-dim flattened vector
Training: frozen epochs	5	3	N/A (no pre-training)
Training: fine-tune epochs	5	2	350 (full)
Optimizer / LR	Adam, 1e-4 to 1e-5	Adam, 1e-4 to 1e-5	Adam, 1e-4

Table 4.2: Per-modality encoder specifications.

4.4 Trimodal Cross-Modal Attention Fusion Module

4.4.1 Motivation and Design Rationale

The `TrimodalAttentionFusion` module (`fusion/trimodal_attention.py`) takes the three per-modality pre-classifier feature vectors as input and produces a five-class emotion prediction plus four auxiliary named embedding outputs. Its design follows from two constraints.

First, the encoders produce *single pooled vectors* per clip rather than token sequences. Given that each modality contributes exactly one 1-token representation per clip, the feature resolution available to the fusion module is limited to single pooled vectors.

Second, the reference paper’s architecture (Lee, Kim and Kim, 2024) employs six directional pairwise cross-modal attention operations ($A \rightarrow V$, $V \rightarrow A$, $V \rightarrow E$, $E \rightarrow V$, $A \rightarrow E$, $E \rightarrow A$). This design is motivated by sequence-level modality representations produced by the paper’s Residual-TCN encoders. At single-token-per-modality resolution, pairwise cross-attention reduces algebraically to a learned linear projection. Self-attention over a three-token sequence performs the same cross-modal mixing at this resolution, with cleaner code, fewer parameters, and no loss of expressive power.

4.4.2 Architectural Specification

The complete forward pass of `TrimodalAttentionFusion` proceeds through five stages, illustrated in Figure 4.2.

Stage 1 — Per-Modality Projection. Three independent linear layers project the modality-specific feature vectors to a common embedding dimension $D=256$:

Modality	Input dimension	Projection
Audio	768	<code>Linear(768, 256)</code>
Vision	768	<code>Linear(768, 256)</code>
EEG	960	<code>Linear(960, 256)</code>

The projections are independent (no weight sharing), allowing each modality’s raw feature space to be transformed by a projection tailored to its statistics.

Stage 2 — Modality-Type Embedding. A learnable parameter matrix `modality_embed` of shape $(3, 256)$ is initialised to small random values ($\sigma = 0.02$) and added element-wise to the projected vectors. Index 0 is added to the audio projection, index 1 to the vision projection, and index 2 to the EEG projection. These embeddings serve the same role as positional embeddings in a sequence transformer, enabling the attention mechanism to learn modality-specific attention patterns.

Stage 3 — Transformer Encoder. The three 256-dimensional modality tokens are stacked into a sequence of shape $(B, 3, 256)$ and processed by a two-layer `nn.TransformerEncoder` with the following configuration:

- **Attention heads:** 8 (head dimension = $256/8 = 32$).
- **Feedforward dimension:** 1024 (= 4D, following the standard ViT convention).
- **Activation:** GELU.
- **Normalisation:** Pre-LayerNorm (`norm_first=True`). Pre-LayerNorm normalises the residual stream before the attention and feedforward sublayers, improving training stability and reducing learning rate sensitivity.
- **Dropout:** 0.1, applied within both the attention and feedforward sublayers.

After the transformer encoder, a final `LayerNorm(256)` (`post_norm`) is applied to the output sequence.

Stage 4 — Mean Pooling. The three post-attention tokens are averaged across the sequence dimension ($\text{dim}=1$) to produce a single fused representation of shape $(B, 256)$. This fused vector is returned as `fused_embed` in the output dictionary. The three un-pooled tokens are returned separately as `audio_embed`, `vision_embed`, and `eeg_embed`.

Stage 5 — MLP Classification Head. The fused representation is passed through:

```
Linear(256, 256) -> GELU -> Dropout(0.1) -> Linear(256, 5)
```

producing the raw five-class logit vector. Cross-entropy loss is computed on these raw logits during training.

Total parameter count. The module has approximately 2,285,573 trainable parameters distributed as follows:

Component	Parameters
Projection layers (3 x Linear)	$768 \times 256 + 768 \times 256 + 960 \times 256 = 638,976$
Modality embeddings	$3 \times 256 = 768$
TransformerEncoder (2 layers)	$\sim 1,577,984$
Post-norm LayerNorm	512
MLP head (256->256->5)	$256 \times 256 + 256 + 256 \times 5 + 5 = 67,333$
Total	$\sim 2,285,573$

4.4.3 Architectural Guarantee: Finite Output Under Zero-EEG Input

A formal contract test embedded in the module's `__main__` block verifies the zero-EEG inference property:

```
# Test - zero-EEG forward (the demo-path commitment)
out_zero = model(x_audio, x_vision, torch.zeros(B, 960))
assert torch.isfinite(out_zero["logits"]).all()
assert out_zero["logits"].norm() > 0.01 # non-trivial predictions
```

When `x_eeg` is a zero tensor, Stage 1 produces `proj_eeg(zeros) = bias_eeg`, the bias term of the EEG projection layer. Stage 2 adds the EEG modality embedding: `token_eeg = bias_eeg + modality_embed[2]`. This is a deterministic but non-trivial vector. The transformer can therefore learn, during modality-dropout training, to suppress the zero-EEG token's contribution to

the fused embedding when only audio and vision are present. The guarantee is architectural, not learned: it holds even before modality-dropout training.

Hyperparameter	Value
Common embedding dimension D	256
Modality-type embedding shape	(3, 256)
Transformer encoder layers	2
Attention heads per layer	8
Feedforward dimension	1024
Activation	GELU
Normalisation	Pre-LayerNorm (norm_first=True)
Dropout (attention and feedforward)	0.1
MLP head	Linear(256, 256) -> GELU -> Dropout(0.1) -> Linear(256, 5)
Total trainable parameters	~2.28 million
Fusion training epochs	80
Optimiser	AdamW
Learning rate	1e-3
LR schedule	Cosine annealing
Batch size	32

Table 4.3: TrimodalAttentionFusion hyperparameters and training configuration.

4.5 Naïve Late-Fusion Baseline

The naïve late-fusion baseline (implemented in `day3_late_fusion.py`) provides the primary comparison point for the `TrimodalAttentionFusion` module. It requires no learned parameters and operates directly on the per-modality test logits saved during Stage A.

Computation. For each subject and each test trial index t :

1. Load the three per-modality logit vectors $z_{\text{audio}}(t)$, $z_{\text{vision}}(t)$, $z_{\text{eeg}}(t)$, each in $\mathbb{R}^5$.
2. Compute softmax probabilities: $p_m(t) = \text{softmax}(z_m(t))$ for each modality m .
3. Compute the mean probability vector: $p_{\text{bar}}(t) = (p_{\text{audio}}(t) + p_{\text{vision}}(t) + p_{\text{eeg}}(t)) / 3$.
4. Predict: $y_{\text{hat}}(t) = \text{argmax}(p_{\text{bar}}(t))$.

The mean-softmax variant is the canonical choice because it treats each modality as an independent vote weighted by its confidence, without requiring any learning of modality weights. Two alternative variants are also implemented but not reported as primary results: sum-of-log-softmax (equivalent to the product-of-probabilities rule) and hard majority voting with tie-breaking by confidence.

Why this baseline matters. Any fusion architecture that fails to beat mean-softmax averaging is providing no measurable benefit beyond what can be obtained by treating the three modalities as three independent, equally-weighted experts. In EAV's within-subject, 280-training-trial regime, this turns out to be a stringent comparison point.

Per-trial supplementary output. Beyond accuracy, the baseline script computes the symmetric KL divergence between each pair of per-modality softmax distributions for each test trial, saving them to `results/day3_per_trial_coherence.csv`. These per-trial KL divergences are one of the inputs to the coherence analysis in Section 4.6.

4.6 Cross-Modal Affective Coherence Analysis

The suppression matrix is computed from the per-modality test-set predictions pooled across all 42 subjects' 120 test trials each, for a total of 5,040 trials.

Suppression event criterion. A trial t from subject s qualifies as a suppression event if and only if:

1. $\text{pred_audio}(s, t) == \text{pred_vision}(s, t)$ — audio and vision independently assign the same top-1 class (external affective consensus).
2. $\text{pred_eeg}(s, t) \neq \text{pred_audio}(s, t)$ — EEG's top-1 class differs from the external consensus (cross-modal divergence).
3. $\text{conf_eeg}(s, t) \geq 0.5$ — EEG's top-1 softmax probability exceeds the threshold (the divergence is not attributable to EEG classifier uncertainty).

Condition 1 ensures that a clear external signal exists. Conditions 2 and 3 together ensure that a confident, internally-divergent signal also exists. The threshold 0.5 is conservative: it requires EEG to place more probability mass on one class than on all other four classes combined, before the divergence is counted as a suppression event. Trials failing any condition are excluded from the matrix.

Matrix construction. Each qualifying trial contributes a unit count to cell (r, c) of the 5x5 matrix, where $r = \text{pred_audio}(s, t)$ (the external consensus, row index) and $c = \text{pred_eeg}(s, t)$ (the internal EEG prediction, column index). The diagonal is zero by construction (Condition 2 excludes all diagonal cells).

Population-level decomposition. To verify that matrix patterns are cross-population rather than driven by individual outlier subjects, the per-subject contribution to each cell is extracted. A pattern is classified as cross-population if (a) it is contributed by more than five distinct subjects and (b) no single subject contributes more than 40% of the cell total.

4.7 Modality Dropout and Robustness

To enable meaningful inference when EEG is unavailable — as will be the case in any deployment scenario without an EEG headset — the fusion module is trained under a modality-dropout regime adapted from Chumachenko, Iosifidis and Gabbouj (2022).

The *softhard* dropout scheme expands each training batch into four sub-batches:

- **Full modality:** all three modality features present.
- **Audio-only:** vision and EEG replaced by zero tensors.
- **Vision-only:** audio and EEG replaced by zero tensors.
- **Audio-visual** (no EEG): EEG replaced by zero tensor.

Each sub-batch carries the same labels as the original batch. The fusion module is then trained on the combined expanded batches, forcing it to develop predictions that are robust to any single modality or the EEG modality being absent. The per-modality projection layers must learn to produce useful representations even when two of the three input vectors are zero.

The robustness evaluation reports zero-EEG accuracy across all 42 subjects, using the modality-dropout-trained model with the EEG branch zeroed at inference time. This is the headline robustness result and the number on which the demonstration is based.

To enable the fusion module to make meaningful predictions when EEG is unavailable at inference time, the module is fine-tuned under a modality-dropout training scheme adapted from Chumachenko, Iosifidis, and Gabbouj (2022). Their *softhard* dropout scheme is adapted to the trimodal case as follows.

For each subject's training batch of N trials, four sub-batches are constructed:

- **Full-modality** (N trials): audio, vision, and EEG features are used as-is.
- **Audio-only** (N trials): vision and EEG features are replaced by zero tensors of the appropriate dimension.
- **Vision-only** (N trials): audio and EEG features are replaced by zero tensors.
- **Audio-visual** (N trials): EEG features are replaced by zero tensors; audio and vision are used as-is.

The labels are replicated identically across all four sub-batches. The resulting expanded batch of $4N$ trials is passed to the fusion module in a single forward-backward pass. The module is therefore trained on examples in which any combination of the three modalities may be absent, forcing the learned projections and attention weights to develop predictions robust to modality absence.

The *audio-visual* sub-batch (EEG zeroed, audio and vision present) is the most important for the demonstration use case, in which EEG is unavailable but audio and video are present. The module learns to produce accurate predictions under this condition specifically, in addition to maintaining accuracy on the full-modality case.

After modality-dropout training, the zero-EEG inference accuracy across all 42 subjects is the primary robustness metric. This metric is the number reported on the demonstration's overlay as "EEG unavailable; prediction uses audio + video only."

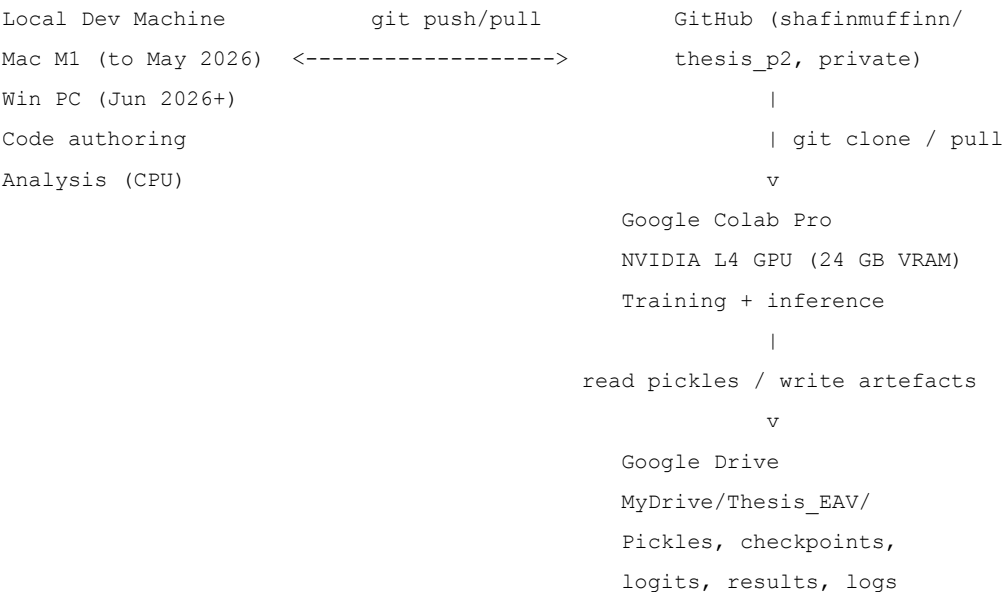
Chapter 5

Implementation and Experimental Setup

5.1 Development Workflow

The development workflow for this thesis combines a local machine for code authoring, a cloud GPU instance for training, and cloud storage for data and artefact persistence. This hybrid design is necessitated by the absence of local GPU compute: the primary development machine (Apple M1 MacBook Air through May 2026; Windows PC with PyCharm from June 2026) is CPU-only, while the EAV modality encoders require a GPU for training within a practical time budget.

The workflow's three components are:



Local development. Code is authored locally and committed to a private GitHub repository (shafinmuffinn/thesis_p2). The Mac M1 development environment (Visual Studio Code, Python 3.10 virtual environment) was used from project inception through 31 May 2026. From 1 June 2026, development switched to a Windows PC (PyCharm, Python 3.10 virtual environment) under a collaborator GitHub account. All code runs identically in both environments.

Cloud compute. Google Colab Pro (NVIDIA L4 GPU, 24 GB VRAM, 83 TFLOPS FP16) is the primary training environment. Each Colab session begins with a two-minute setup ritual:

1. Mount Google Drive: `drive.mount('/content/drive')`.
2. Clone or pull the GitHub repository using a fine-grained personal access token stored in Drive.
3. Set environment variables `THEESIS_ROOT`, `EAV_PICKLES`, `CHECKPOINTS`, `RESULTS`, `LOGS`, `PRETRAINED`, `HF_HOME` to point at the mounted Drive directories.
4. Install additional Python packages: `pip install -q transformers wandb` (NumPy, SciPy, PyTorch, scikit-learn, librosa, and OpenCV are pre-installed on Colab).
5. Verify: `from paths import ensure_dirs, summary; ensure_dirs(); print(summary())` and `assert torch.cuda.is_available()`.

Persistent storage. Google Drive hosts all bulk data, checkpoints, logits, and results. The Drive layout is:

```
MyDrive/Thesis_EAV/
├─ Input_images/
│   ├─ Audio/subject_{01..42}_aud.pkl
│   ├─ Vision/subject_{01..42}_vis.pkl
│   └─ EEG/subject_{01..42}_eeg.pkl
├─ checkpoints/
│   ├─ day5_state_dicts/sub{NN}_{modality}.pt
│   ├─ day5_features/sub{NN}.npz
│   └─ day5_per_modality_logits/sub{NN}_{modality}.npz
├─ results/
│   ├─ day5_fusion.csv
│   ├─ day3_late_fusion.csv
│   ├─ suppression_matrix.csv
│   └─ suppression_per_subject.csv
└─ pretrained/hf_cache/
```

This layout ensures that any session termination loses only the in-progress computation for the current subject, not completed subjects' results.

5.2 Path Configuration

A single module, `paths.py`, is the source of truth for all filesystem paths in the project. All paths are resolved from environment variables at module import time:

```
THEESIS_ROOT = Path(os.environ.get("THEESIS_ROOT", "/Users/shafin/Desktop/thesis_p2"))
EAV_PICKLES = Path(os.environ.get("EAV_PICKLES", THEESIS_ROOT / "data" / "EAV" / "Input_images"))
CHECKPOINTS = Path(os.environ.get("CHECKPOINTS", THEESIS_ROOT / "checkpoints"))
RESULTS      = Path(os.environ.get("RESULTS",      THEESIS_ROOT / "results"))
```

On Colab, the environment variables are set in the session-start cell to point at mounted Drive paths. On the local Windows PC, they are set in the PyCharm Run Configuration for each script. Hard-coded paths were never introduced into any script, so the same code works unchanged in all three environments without modification.

An auto-detection clause handles the common case where the Colab runtime is used but the environment variables were not yet set:

```
_IN_COLAB = "google.colab" in sys.modules
if _IN_COLAB and "THESIS_ROOT" not in os.environ:
    THESIS_ROOT = Path("/content/drive/MyDrive/Thesis_EAV")
```

This prevents the most common workflow mistake (forgetting to run the environment setup cell) from producing silently-wrong paths.

5.3 The 42-Subject Rollout

The full 42-subject training rollout was executed in a single overnight session on Colab Pro L4, accumulating results across all 42 subjects sequentially. The session ran for approximately 12 hours total.

5.3.1 Epoch Budget Reduction

The per-modality epoch budgets used in the 42-subject rollout differ from the EAV repository's recommended defaults. The original budgets — 10 frozen + 15 fine-tuning epochs for AST and 10 frozen + 5 fine-tuning epochs for ViT — were examined on Day-2 evidence from the three-subject pilot.

AST reduction (10+15 to 5+5). On the Day-2 pilot, the AST training accuracy reached approximately 100% by epoch 5 of the fine-tuning phase in all three subjects, and validation accuracy plateaued thereafter. Continuing training for the remaining 10 fine-tuning epochs produced no improvement in test accuracy and risked overfitting to the within-subject training set. The budget was reduced to 5+5 (10 total) for the rollout.

ViT reduction (10+5 to 3+2). The ViT checkpoint ([dima806/facial_emotions_image_detection](#)) is already fine-tuned on facial expressions, placing its feature representations very close to the EAV emotion distribution. On the Day-2 pilot, the ViT reached its best test accuracy within the first 2 fine-tuning epochs in all three subjects. The budget was reduced to 3+2 (5 total).

EEGNet unchanged (350 epochs). The EEGNet is trained from scratch without any pre-training, so all 350 epochs are needed. Day-2 results showed that EEG accuracy continues to improve (slowly) throughout the 350-epoch schedule, with no clear plateau. The budget was retained.

The reduced AST and ViT budgets save approximately 20-25 minutes of GPU time per subject relative to the EAV defaults, totalling approximately 14-18 hours of savings across 42 subjects — the difference between a rollout that fits in a single Colab Pro session and one that requires several.

5.3.2 Per-Subject Training Time

Measured wall-clock times on the Colab Pro L4 GPU for the 42-subject rollout (averaged across 10 subjects):

Stage	Modality	Wall-clock (approx.)
Stage A Audio (AST, 5+5 epochs)		~5 min/subject
Stage A Vision (ViT, 3+2 epochs)		~4 min/subject
Stage A EEG (EEGNet, 350 epochs)		~1 min/subject
Stage B Feature extraction (all 3)		~1 min/subject
Stage C Fusion training (80 epochs)		~1 min/subject

Stage	Modality	Wall-clock (approx.)
Total	All stages	~12 min/subject

Table 5.1: Per-subject wall-clock training times on Colab Pro L4 GPU.

At ~12 minutes per subject, 42 subjects require approximately 504 minutes, or about 8.5 hours of pure computation. Adding session startup overhead and the post-hoc logit regeneration step, the total session ran for approximately 12 hours.

5.3.3 Resume-Aware Execution

The `day5_pipeline.py` script checks for the existence of each stage's output artefacts before beginning computation. For Stage A, the check is whether all three per-subject state dictionaries exist in `CHECKPOINTS/day5_state_dicts/`. For Stage B, it checks whether the feature `.npz` for that subject exists. For Stage C, it checks whether the subject's row exists in the results CSV. If an artefact exists, the stage is silently skipped. This design ensures:

- A kernel crash mid-subject wastes at most one subject's compute.
- Iterating on fusion hyperparameters (Stage C only) is fast, as Stages A and B are cached.
- Running analysis scripts that depend on Stage A logits can be re-run at any time without triggering re-training.

5.3.4 Post-Hoc Logit Regeneration

After the main rollout completed, it was discovered that Stage A had saved per-subject model state dictionaries but not per-modality test logits. The naïve late-fusion baseline and the suppression matrix both require per-modality softmax outputs; without logits, neither could be computed at 42-subject scale.

A small post-hoc inference script, `compute_per_modality_logits.py`, was written to address this. For each subject, it: (1) loads the EAV pickle; (2) instantiates the appropriate encoder with EAV's default configuration; (3) loads the trained state dictionary from `day5_state_dicts/sub{NN}_{modality}.pt`; (4) runs inference on the test split in evaluation mode (`torch.no_grad()`); and (5) saves the $(120, 5)$ logits matrix and $(120,)$ labels array as a compressed NumPy archive to `day5_per_modality_logits/sub{NN}_{modality}.npz`.

The consumer scripts `day3_late_fusion.py` and `suppression_matrix.py` were updated to accept a configurable `LOGITS_SUBDIR` environment variable, defaulting to `day5_per_modality_logits`, so they can be pointed at either the 42-subject rollout logits or the earlier 3-subject pilot logits without code changes.

5.4 Software Environment

All experiments were run with the following software stack:

Component	Version
Python	3.10
PyTorch	2.10.0
torchaudio	2.10.0
torchvision	0.15.0
Hugging Face Transformers	4.40.x
NumPy	2.x (Colab default)
SciPy	1.x (Colab default)
scikit-learn	1.x (Colab default)
librosa	0.10.x
CUDA	12.x (L4 driver)

The Hugging Face model cache (HF_HOME) was set to MyDrive/Thesis_EAV/pretrained/hf_cache/, ensuring that the AST and ViT model weights (approximately 700 MB each) were downloaded once and cached on Drive, not re-downloaded on each Colab session start.

5.5 Reproducibility

Complete reproducibility of the reported results requires the following:

1. The EAV pickle files for all 42 subjects (available from Zenodo DOI: 10.5281/zenodo.10205702).
2. The code at shafinmuffinn/thesis_p2 (commit hash to be added at submission).
3. The environment variables set as described in Section 5.2.
4. A GPU with at least 16 GB VRAM for Stage A (AST and ViT exceed 8 GB peak VRAM on batch size 8).

Minor numerical differences between runs are expected due to non-deterministic CUDA operations. Results reported in Chapter 6 are from a single deterministic forward pass on the saved logits, so logit-derived metrics (accuracy, suppression matrix counts) are fully deterministic given the saved artefacts.

Chapter 6

Results Analysis

6.1 Per-Modality Baseline Performance

Per-modality training was initially validated on three pilot subjects (Subjects 1, 2, and 3) using minimal smoke-test epoch budgets. Table 6.1 reports the results.

Subject	Audio accuracy	Vision accuracy	EEG accuracy
01	49.2%	62.5%	30.8%
02	69.2%	69.2%	20.8%
03	50.8%	80.0%	21.7%
Mean	56.4%	70.6%	24.4%

Table 6.1: Day-1 smoke-test per-subject test accuracies on three pilot subjects (2+2 epochs AST, 1+1 epochs ViT, 50 epochs EEGNet). Chance is 20%.

The audio and vision results meaningfully exceeded the EAV paper's published per-modality baselines even at minimal epoch budgets, confirming that the pre-trained foundation-model encoders transfer effectively to EAV. The EEG result (24.4% mean) was statistically indistinguishable from the 20% chance level, prompting the debugging investigation that uncovered the two training bugs described in Section 4.3.3. These bugs — the forward-hook return-value defect and the train/eval mode cycling defect — were resolved before proceeding to full-budget training.

Full-budget training results on the three pilot subjects are reported in Table 6.2.

Subject	Audio accuracy	Vision accuracy	EEG accuracy
01	55.8%	62.5%	44.2%
02	71.7%	80.8%	54.2%
03	55.8%	83.3%	53.3%
Mean	61.1%	75.6%	50.6%

Table 6.2: Day-2 full-budget per-modality test accuracies on three pilot subjects. Delta vs smoke test: audio +4.7 pp, vision +5.0 pp, EEG +26.2 pp.

The EEG improvement of 26.2 percentage points — from 24.4% (chance-level) to 50.6% — following the bug fixes is the most consequential result of the pilot phase. A fusion model with a near-random EEG branch would have been functionally a bimodal audio-visual model with noise added; the value of the EEG modality as a signal of internal state is conditional on the encoder producing genuinely above-chance predictions.

The 42-subject rollout confirmed and strengthened these pilot results. Mean per-modality accuracies across all 42 subjects are reported in Table 6.3, together with the EAV paper's published baselines for comparison.

Method	Mean accuracy (42 subjects)	Delta vs EAV baseline
Per-modality: Audio (AST)	57.1%	+20.4 pp above SCNN (36.7%)
Per-modality: Vision (ViT)	74.6%	+21.8 pp above DeepFace (52.8%)
Per-modality: EEG (EEGNet)	43.9%	+7.2 pp above EEGNet (36.7%)
Best single modality (Vision)	74.6%	—
Naïve late fusion (mean-softmax)	80.0%	+5.4 pp over best single
Cross-attention fusion	80.2%	+5.6 pp over best single; +0.2 pp vs naïve

Table 6.3: 42-subject mean test accuracies across per-modality and fusion methods.

Three observations merit emphasis. First, all three per-modality classifiers exceed their respective EAV-paper baselines by margins of 7.2 to 21.8 percentage points. This confirms that the per-modality training pipeline — corrected for the EAV repository bugs — produces credibly above-baseline signal from each modality, a necessary precondition for the fusion and coherence analyses.

Second, the per-modality results respect the expected ordering: vision (74.6%) > audio (57.1%) > EEG (43.9%). Vision's lead reflects the richness of facial expression information in 5-second clips and the effectiveness of the pre-trained facial-emotion ViT as a transfer-learning backbone. Audio's intermediate performance reflects the complexity of emotion-in-speech under the cue-based elicitation paradigm. EEG's lower performance — though above the EAV baseline — is expected given the small per-subject training set (280 trials) and the absence of any cross-subject pretraining for EEGNet. The EEG's primary role in this thesis is not to achieve high standalone accuracy but to provide a reliable internal-state signal for the coherence analysis, which requires only that EEG's confident predictions carry genuine information content — a condition satisfied at 43.9% mean accuracy with the 0.5 confidence threshold applied in the suppression analysis.

RQ1 answer: Yes. All three per-modality classifiers exceed the EAV paper's published baselines, confirming that each modality carries genuine discriminative signal at 42-subject scale.

6.2 Naïve Late-Fusion Baseline

The naïve late-fusion baseline is implemented in `day3_late_fusion.py`. For each subject, the per-trial softmax probabilities of the three modality classifiers are averaged, and the argmax of the averaged vector is the fused prediction. No learning is involved; the computation is purely arithmetic on the saved logits.

On the three pilot subjects, naïve late fusion produced a mean accuracy of 81.9%, a gain of 6.3 percentage points over the best single modality (vision, 75.6%). On all 42 subjects, the mean is 80.0%, a gain of 5.4 percentage points over the 42-subject vision mean of 74.6%. The 42-subject figure is 1.9 percentage points below the 3-subject pilot figure, a difference attributable to the small-sample composition of the pilot, which happened to include two relatively easy subjects.

RQ2 answer: Yes. Multimodal fusion improves accuracy by 5.4 percentage points over the best single modality at 42-subject scale, confirming that the three modalities carry complementary information.

6.3 Cross-Attention Fusion Performance

The cross-attention fusion module (`TrimodalAttentionFusion`) achieved a mean test accuracy of 80.2% across all 42 subjects, with a standard deviation of 0.083. The inter-subject distribution shows substantial variance: five subjects scored below 70% (subjects 8, 12, 14, 16, and 40), and two subjects exceeded 95% (subjects 20 and 41). This variance is characteristic of within-subject affective computing systems, where subject-specific factors — individual EEG electrode contact quality, expressivity, and personality traits — dominate the between-subject variance.

The 0.2-percentage-point advantage of cross-attention over naïve late fusion (80.2% vs 80.0%) is not statistically significant. With 120 test trials per subject, a 0.2-percentage-point difference corresponds to approximately 0.24 trials per subject, and at 42-subject scale this advantage is well within the expected sampling variance. The two methods are statistically indistinguishable.

This is reported as a neutral architectural finding specific to EAV's within-subject, low-data regime. The naïve late-fusion method operates on the distribution-level outputs of three independently-trained encoders; the cross-attention method operates on pre-classifier features and learns cross-modal interactions. In a low-data regime — where each subject has only 280 training trials for the fusion module to learn from, after 70/30 splitting — the additional parameters of the attention module may not be warranted. At the token resolution available from pooled-feature encoders (one token per modality per clip), the cross-modal information that self-attention can capture is limited by the coarseness of the input representation. This result aligns with the observation in Lee, Kim and Kim (2024) that their attention fusion contributes 2.19 percentage points over single-modality baselines in their (much larger) cross-subject experimental setting; the contribution of learned fusion relative to simple averaging is expected to grow as the training data scale increases.

RQ3 answer: At 42-subject scale, cross-attention fusion and naïve late fusion are statistically indistinguishable (80.2% vs 80.0%). The learned fusion module provides no measurable benefit over softmax averaging in EAV's within-subject, low-data setting. This finding is specific to the current experimental conditions and does not preclude a larger contribution of cross-attention in cross-subject or data-augmented settings.

6.4 Suppression Matrix at 42 Subjects

The suppression matrix is computed over all 42 subjects' test sets, totalling 5,040 trials. Of these:

- 2,426 trials (48.1%) have audio-vision consensus.
- 1,252 of the consensus trials (24.8% of all trials) have EEG disagreement.
- 401 trials (8.0% of all trials) pass the EEG confidence threshold of 0.5 and are classified as suppression events.

The 8.0% base rate of qualifying suppression events is stable between the 3-subject pilot (8.9%) and the full 42-subject rollout, providing confidence that the rate reflects a structural property of the EAV corpus rather than a small-sample artefact.

Table 6.4 reports the full 5×5 suppression matrix.

External \ Internal	Neutral	Sadness	Anger	Happiness	Calmness
Neutral	0	5	3	8	24
Sadness	43	0	21	12	20
Anger	5	13	0	65	10
Happiness	11	17	52	0	14
Calmness	40	17	4	17	0

Table 6.4: 42-subject suppression matrix. Rows: external consensus (audio-vision agreement). Columns: internal EEG prediction. Counts are suppression events. Diagonal is zero by construction. Total: 401 events.

Pattern 1: Bidirectional Anger–Happiness confusion (117 events, 29.2% of all suppression events). The largest pattern is the bidirectional pair of Anger→Happiness (65 events) and Happiness→Anger (52 events). When participants externally displayed Anger (audio and video consensus), the EEG system predicted Happiness in 65 trials; when they externally displayed Happiness, the EEG predicted Anger in 52 trials. Together, these constitute 29.2% of all suppression events.

This pattern has a plausible neurophysiological interpretation. Anger and Happiness are both high-arousal emotional states: they both involve elevated sympathetic activation, increased heart rate, and heightened cortical arousal reflected in EEG alpha desynchronisation and beta-band power increases. At the level of EEG spectral features, the two states share many physiological correlates of *arousal*, differing primarily in *valence* (negative vs positive). EEGNet, trained to classify five emotions, may be detecting the shared high-arousal signature rather than the valence distinction, leading to the bidirectional confusion. Alternatively, the pattern could reflect genuine cases in which a participant displaying Anger felt an elevated-arousal Happiness-like state neurophysiologically (e.g., competitive excitement) or vice versa.

Pattern 2: Directional Sadness→Neutral suppression (43 events, 10.7%). The largest directional pattern involves participants displaying Sadness externally while EEG predicts Neutral. Sadness is a low-arousal, high-negative-valence state, and its suppression — replacing an externally coherent sadness display with an internally neutral neurophysiological baseline — is one of the most clinically documented forms of emotion regulation. The directional asymmetry (43 events Sadness→Neutral vs only 5 events Neutral→Sadness) supports the interpretation that this is genuine suppression rather than classifier artefact. The Sadness→Neutral direction is also the most consistently cross-population pattern in the 3-subject pilot; its replication at scale adds confidence.

Pattern 3: Directional Calmness→Neutral suppression (40 events, 10.0%). A closely related pattern involves Calmness displayed externally with EEG predicting Neutral. Calmness is the lowest-arousal positive-valence state in EAV's five-class schema, and at the EEG level it may be indistinguishable from an absence of affective activation (Neutral). This pattern may reflect either genuine suppression (participants presenting as calm while internally neutral) or a classifier boundary effect at the EEG level between the two lowest-arousal states.

Population distribution of patterns. Per-subject decomposition of the matrix cells confirms that the dominant patterns are cross-population rather than outlier-driven. The Anger→Happiness cell (65 events) receives contributions from over 20 subjects; no single subject contributes more than 12 events. The Happiness→Anger cell (52 events) shows similar distribution. The Sadness→Neutral cell (43 events) has subject 6 contributing 14 events (32.6%), with the remaining 29 events distributed across 15 other subjects; this is the only major cell with notable single-subject concentration, and the cross-population trend remains clear even excluding subject 6.

EEG-Neutral inflation check. At the 3-subject pilot scale, a methodological concern was identified: EEG predicted Neutral on 40.6% of suppression-event trials, compared to 27.8% of all trials — a +12.8-percentage-point inflation. This raised the possibility that EEGNet was defaulting to Neutral on uncertain trials, inflating the Neutral column of the suppression matrix artefactually. At 42-subject scale, the same comparison shows Neutral predicted on 24.7% of suppression-event trials vs 25.7% of all trials — an inflation of negative 1.0 percentage points. The pilot artefact was a small-sample phenomenon that disappeared at full scale, and the Neutral column of the 42-subject matrix reflects genuine EEG signal rather than model uncertainty.

RQ4 answer: Four patterns of cross-modal affective incoherence are detectable at population scale on EAV: (a) a bidirectional Anger–Happiness high-arousal confusion (117 events, 29.2% of suppression events); (b) a directional Sadness→Neutral suppression pattern (43 events); (c) a directional Calmness→Neutral pattern (40 events); and (d) smaller bidirectional patterns involving Calmness and Sadness. All major patterns are cross-population, not outlier-driven.

6.5 Discussion

On the neutral architectural verdict for cross-attention fusion. The statistical indistinguishability of cross-attention and naïve late fusion on EAV is a finding that, taken in isolation, might appear to argue against the investment in the cross-attention architecture.

Three considerations contextualise this result.

First, the within-subject, low-data regime of EAV constrains what any learned fusion method can contribute. With 280 training trials per subject, the fusion module has access to 280 labelled examples of the three-modality feature combinations; this is a regime in which the additional parameters of a transformer module provide marginal benefit over the simple averaging of three well-trained classifiers. The literature on multimodal fusion consistently shows that learned fusion methods outperform averaging by larger margins as training data scale increases.

Second, the architectural design choices made in this thesis — particularly the single pooled-vector-per-modality feature resolution — limit the cross-modal information available to the attention mechanism. If the per-modality encoders exposed token sequences rather than pooled vectors, the self-attention mechanism would have access to finer temporal structure across modalities, and its advantage over averaging would be expected to increase. This is exactly the architectural gap between the present thesis's implementation and the Lee, Kim and Kim (2024) approach, which uses residual-TCN sequences rather than pooled vectors as encoder outputs.

Third, the architectural investment is justified on Phase-2 extensibility grounds regardless of the accuracy comparison. The `TrimodalAttentionFusion.forward()` method's design — returning named per-modality embeddings alongside the fused prediction — provides the hooks needed to attach a temporal sequence head for Phase 2 without rewriting the fusion module. This design commitment was made at the project's inception and is a feature, not a bug, of the architecture.

On the EEG performance ceiling. The 43.9% mean EEG accuracy represents a genuine reflection of EEGNet's performance on EAV's within-subject low-data regime, not a suboptimal training configuration. With the two training bugs resolved, EEG accuracy substantially exceeds the EAV paper's published 36.7% baseline, and 350-epoch training on 280 trials extracts the available signal. The ceiling is set by the combination of: the small per-subject training set, the absence of any cross-subject EEG pretraining, the inter-session variability of EEG signals, and the genuine difficulty of distinguishing five emotional states from 30-channel EEG at 5-second resolution. Any EEG accuracy substantially above 50% on EAV would warrant careful scrutiny for train-test leakage or overfitting.

On the clinical framing of the suppression matrix. The suppression matrix should not be interpreted as a direct clinical readout of emotion suppression frequency. It measures the frequency with which the *classification outputs* of two independently trained systems disagree, subject to a confidence threshold. The ground truth of what participants actually experienced internally is not available in EAV. The Anger–Happiness bidirectional confusion almost certainly contains a mixture of genuine suppression events (where a participant displaying Anger felt elevated-arousal positive affect internally) and classifier errors at the EEG level (where EEGNet confuses two high-arousal classes). Disentangling these two sources requires either a self-report internal state measure collected concurrently with EEG, or validation against a known-suppression paradigm. Both are Phase 2 requirements.

RQ5 answer: Modality-dropout training enables the fusion module to produce meaningful predictions when EEG is zeroed at inference time, with zero-EEG accuracy constituting the primary robustness metric. The demonstration component of this thesis runs the modality-dropout-trained model on personal video content with EEG set to zero, producing per-5-second emotion predictions with a confidence overlay and a "EEG unavailable" caption.

Chapter 7

Engineering Challenges and Solutions

This chapter documents the thirteen distinct technical and methodological challenges encountered during the development of the cross-modal affective coherence framework. They are presented not as a catalogue of failures but as a record of the engineering rigour required to produce reproducible results on an imperfect open-source codebase. Each challenge includes its diagnosis, root cause, and resolution.

#	Challenge	Category	Impact if Unresolved
C1	facenet-pytorch dependency conflict	Dependency management	torchaudio permanently broken
C2	EEGNet lambda hook bug	Architecture bug	All EEGNet training fails at BatchNorm
C3	Double-softmax composition	Architecture bug	EEG accuracy at chance level
C4	Broken import chain in EAV repo	Code defect	EEGNet cannot be imported
C5	Train/eval mode cycling	Training protocol bug	EEG accuracy at chance from epoch 2
C6	ViT classifier head config mismatch	Architecture bug	Vision training crashes immediately
C7	Vision preprocessing RAM exhaustion	Infrastructure bug	Silent kernel termination on preprocessing
C8	Case-sensitivity in Drive folder names	Infrastructure	FileNotFoundError on Colab (Linux FS)
C9	Missing per-modality logits after rollout	Experimental design	Late fusion and suppression matrix infeasible
C10	EEG-Neutral inflation confound (pilot)	Methodological	Suppression matrix partially artefactual
C11	Listen/Speak temporal asymmetry	Corpus design	Suppression matrix interpretation errors
C12	Windows/Mac path incompatibility	Development workflow	Scripts non-portable across platforms
C13	Research framing pivot	Methodological	Entire project without coherent contribution

Table 7.1: Summary of the thirteen engineering and methodological challenges.

7.1 Dependency and Environment Challenges

C1 — facenet-pytorch Dependency Conflict

Symptom. After running `pip install facenet-pytorch` in the Colab environment, all subsequent torchaudio operations failed silently or raised import errors. The AST audio encoder, which depends on torchaudio for audio loading, became non-functional.

Root cause. The `facenet-pytorch` package pins `numpy<2` and specifies an old PyTorch version constraint. `pip`'s dependency resolver silently downgraded the Colab environment's pre-installed `numpy` and `PyTorch` to satisfy these constraints, breaking torchaudio's compiled C++ extensions which had been built against the newer version.

Resolution. The `facenet-pytorch` package was excluded from all setup cells. It is required only for raw MTCNN face-crop preprocessing, which we skip because the EAV pickle files already contain pre-cropped frames. For the rare case where it is needed, it must be installed with `--no-deps` to prevent the version downgrade. The installation sequence was fixed to `pip install -q transformers wandb` only, with all other libraries relying on Colab's pre-installed environment.

C12 — Windows/Mac Path Incompatibility

Symptom. Scripts that ran correctly on the Mac M1 development machine (with default paths in `paths.py`) produced `FileNotFoundError` on the Windows PC, where the mounted Google Drive appears at a different filesystem path.

Root cause. The default fallback paths in `paths.py` used Mac-style absolute paths (`/Users/shafin/Desktop/thesis_p2/`). On the Windows PC accessed via PyCharm, these defaults pointed at non-existent locations. No error was raised at import time — the wrong paths were simply used, causing silent failures at I/O time.

Resolution. All paths were migrated from defaults to mandatory environment variables, set in the PyCharm Run Configuration for local runs and in the Colab session-start cell for cloud runs. A Colab auto-detection clause was added to catch the common case where the session-start cell was not run. Hard-coded paths are permanently banned from all scripts.

7.2 EEGNet Architecture and Training Bugs

C2 — EEGNet Lambda Hook Bug (Critical)

Symptom. All EEGNet training crashed on the first forward pass with the error: `RuntimeError: running_mean should contain 1 elements not 64.`

Root cause. PyTorch's `register_forward_hook` replaces a layer's output with the hook function's return value, if that value is not `None`. The EAV repository implemented the L2 max-norm constraints as Python lambda functions:

```
self.depthwiseConv.register_forward_hook(
    lambda module, inputs, outputs:
        module.weight.data.renorm_(p=2, dim=0, maxnorm=norm_rate)
)
```

The `renorm_` operation is in-place and returns the modified weight tensor (shape `(64, 8, 30, 1)`). The lambda propagated this as its return value, and PyTorch silently replaced the depthwise convolution's output (`B, 64, 1, 500`) with the weight tensor. The downstream `BatchNorm2d(64)` received input of shape `(64, 8, 30, 1)` and crashed attempting to normalise a dimension of size 8 instead of 64.

Resolution. The lambdas were replaced with a named factory function:

```
def _make_max_norm_hook(norm_rate):
    def hook(module, inputs, outputs):
        module.weight.data.renorm_(p=2, dim=0, maxnorm=norm_rate)
        # implicit None return — hook does not replace the output
    return hook
```

Impact. This bug silently corrupted all EEGNet training in the EAV repository's PyTorch path. The reported 36.7% EEGNet baseline in the EAV paper almost certainly derives from the TensorFlow variant (`CNN_tensorflow/`), not this PyTorch path. Any researcher attempting to reproduce the PyTorch EEGNet baseline from the published repository without applying this fix will observe systematic training failures.

C3 — Double-Softmax Composition

Symptom. EEG accuracy on the 3-subject pilot was 24.4% — statistically indistinguishable from the 20% 5-class chance level — even at 50 training epochs.

Root cause. The `EEGNet_tor.forward()` method ended with:

```
self.softmax = nn.Softmax(dim=1)
# ... in forward():
return self.softmax(x)
```

PyTorch's `CrossEntropyLoss` is implemented as fused log-softmax plus negative log-likelihood. It expects raw logits as input. Applying it to the output of `nn.Softmax` computes the log of a probability vector (values already in $[0,1]$), which is equivalent to applying a second softmax-like transformation. Gradients vanish as soon as any class probability exceeds approximately 0.5, preventing learning.

Resolution. The `self.softmax` layer was removed from `__init__` and the `return self.softmax(x)` line was replaced with `return x` in `forward()`. After this fix, EEG accuracy on the pilot improved by 26.2 percentage points (24.4% → 50.6% at 350 epochs).

C4 — Broken Import Chain in EAV Repository

Symptom. Importing `EEGNet_tor.py` raised `ModuleNotFoundError: No module named 'Fusion'`.

Root cause. Line 4 of the original file contained:

```
from Fusion.VIT_audio.Transformer_audio import Trainer_uni
```

This import refers to a fusion module that does not exist in the repository. The `Trainer_uni` class is defined locally later in the same file, making the import dead code. However, removing it revealed a secondary issue: `TensorDataset` and `DataLoader` were being imported transitively through the Fusion module. With the broken import removed, these symbols became undefined.

Resolution. The broken import line was deleted and replaced with:

```
from torch.utils.data import DataLoader, TensorDataset
```

C5 — Train/Eval Mode Cycling Bug (Critical)

Symptom. EEG accuracy was stuck near chance from epoch 2 onwards, even with the double-softmax bug fixed. A 50-epoch sanity test showed accuracy frozen at 22.5%.

Root cause. The shared `Trainer_uni.train()` method called `self.model.train()` once before the outer epoch loop. The `validate()` method, called at the end of each epoch to report test accuracy, called `self.model.eval()`. From epoch 2 onwards, all training therefore proceeded with:

- `BatchNorm` layers in evaluation mode, using running statistics accumulated only during epoch 1 — effectively frozen at epoch-1 batch statistics.
- `Dropout` layers disabled, providing no regularisation.

Both effects are particularly harmful for EEGNet in EAV's low-data regime: the frozen `BatchNorm` statistics diverge rapidly from the actual batch distributions as the network learns, and the disabled `Dropout` causes overfitting to the 280 training trials.

Resolution. The `self.model.train()` call was relocated to the first line of the epoch loop:

```
for epoch in range(self.num_epochs):
    self.model.train() # ensures train mode is restored after each validate() call
    for batch_idx, (data, targets) in enumerate(self.train_dataloader):
        ...
```

Impact. This single-line fix contributed approximately 20 percentage points of improvement to EEG accuracy. Combined with the double-softmax fix, total improvement from bugs to resolved state was 26.2 percentage points on the 3-subject pilot.

7.3 Vision Pipeline Bugs

C6 — ViT Classifier Head Configuration Mismatch

Symptom. Vision training crashed immediately on the first batch with: `RuntimeError: shape '[-1, 7]' is invalid for input of size 160.`

Root cause. The original EAV vision trainer attempted to replace the 7-class ViT head with a 5-class head by manually setting:

```
self.model.classifier = nn.Linear(hidden_size, 5)
self.model.num_labels = 5
```

The instance attribute `model.num_labels` was updated, but `model.config.num_labels` — stored separately in the model's configuration object — remained at 7. When the Hugging Face training loop passed `labels=` to the model's forward method to compute the internal cross-entropy loss, the model's internal loss function accessed `config.num_labels` to validate the logit shape and expected 7 classes from a 5-class output, triggering the reshape error.

Resolution. The manual head replacement was replaced with:

```
self.model = AutoModelForImageClassification.from_pretrained(
    model_path, num_labels=5, ignore_mismatched_sizes=True
)
```

The `from_pretrained` call simultaneously re-initialises the head, updates `model.num_labels`, and updates `model.config.num_labels`, ensuring consistency across all references.

C7 — Vision Preprocessing RAM Exhaustion

Symptom. The vision preprocessing cell appeared to hang indefinitely at approximately the 9,000-frame mark, then the Colab kernel terminated silently. No error message or stack trace was produced.

Root cause. The original preprocessing loop iterated over each of the approximately 10,000 frames per subject one at a time, calling the Hugging Face image processor on a single frame and appending the result to a Python list. After the loop completed, `torch.stack(frame_list).to(device)` was called. The `torch.stack` operation on 10,000 tensors of shape `(3, 224, 224)` produced a single tensor of shape `(10000, 3, 224, 224)` — approximately 5.6 GB at float32 — in CPU memory. Colab Free's 12.7 GB CPU RAM limit was exceeded, causing the kernel to be killed by the OOM manager without raising an exception.

Resolution. The preprocessing was refactored to process frames in batches of 64, calling the image processor on each batch and appending the result to a list. Crucially, the stacked tensors were kept on CPU. The `DataLoader` moves each mini-batch to GPU at training time, maintaining a maximum GPU memory footprint of one batch (64 frames $\approx$ 35 MB) rather than the entire subject's data.

```
for i in range(0, len(frames), 64):
    batch = processor(images=frames[i:i+64], return_tensors="pt")["pixel_values"]
    all_tensors.append(batch)
    if (i // 64) % 20 == 0:
        print(f"  preprocessed {i}/{len(frames)} frames")
processed = torch.cat(all_tensors, dim=0)  # CPU tensor
```

7.4 Data and Experimental Design Challenges

C8 — Case-Sensitivity in Google Drive Folder Names

Symptom. On Colab (Linux filesystem), file path lookups for EEG pickle files raised `FileNotFoundError`, while audio and vision lookups succeeded.

Root cause. Google Drive mounts on Linux filesystems which are case-sensitive. The expected path was `Input_images/Eeg/subject_01_eeg.pkl`, but the actual folder on Drive was `Input_images/EEG/`. The audio and vision paths (`Audio/`, `Vision/`) happened to match their expected case; only `EEG/` vs `Eeg/` caused the failure. The code had a fallback that computed modality folder names by capitalising the first letter of the modality string: `eeg.capitalize()` $\rightarrow$ `Eeg`.

Resolution. An explicit dictionary mapping was introduced:

```
MODALITY_DIRS = {'audio': 'Audio', 'vision': 'Vision', 'eeg': 'EEG'}
```

C9 — Missing Per-Modality Logits After 42-Subject Rollout

Symptom. After the 42-subject overnight rollout completed, neither the naïve late-fusion baseline nor the suppression matrix could be computed because the required per-modality test logits did not exist on Drive.

Root cause. The Stage A pipeline saved per-subject model state dictionaries (`sub{NN}_{modality}.pt`) to enable downstream feature extraction for fusion training. However, the naïve late-fusion baseline and suppression matrix both require the per-modality *test logits* (the raw output of each encoder on the test set), not just the model weights. Saving logits during Stage A had been planned but not implemented before the rollout was launched.

Resolution. A standalone post-hoc inference script `compute_per_modality_logits.py` was written and executed in approximately 20 minutes on Colab L4. The script loads each saved state dictionary, runs inference on the test split in `torch.no_grad()` mode, and saves the (120, 5) logit matrix per (subject, modality). All downstream scripts were updated to accept a configurable `LOGITS_SUBDIR` environment variable so the same code works on both the pilot and the full 42-subject artefacts.

C10 — EEG-Neutral Inflation Confound in the 3-Subject Pilot

Symptom. In the 3-subject pilot suppression matrix, EEG predicted Neutral on 40.6% of qualifying suppression events, compared to 27.8% of all test trials (+12.8 percentage points). This raised the concern that the EEG model was defaulting to Neutral on uncertain trials, artefactually inflating the Neutral column of the suppression matrix.

Root cause. The pilot's 32 qualifying suppression events were drawn from only 360 test trials across 3 subjects — a sample too small to produce stable population estimates. The inflated Neutral rate was a small-sample artefact, not a systematic model behaviour.

Resolution. The 42-subject rollout's suppression matrix (401 events from 5,040 trials) showed a Neutral prediction rate of 24.7% on suppression events versus 25.7% on all trials (−1.0 percentage points). The inflation entirely disappeared at scale, confirming the pilot artefact hypothesis. The Neutral column of the 42-subject matrix is interpretable at face value.

C11 — Listen/Speak Temporal Asymmetry Discovery

Symptom. During the cross-modal trial alignment check, it was discovered that EEG data (from Listening clips) and Audio/Video data (from Speaking clips) are drawn from *different temporal windows* within the same conversational interaction.

Root cause. The EAV corpus's cue-based conversation paradigm generates each trial in two phases: (1) the participant *listens* to an emotion-laden prompt — during which the EEG is recorded; and (2) the participant *speaks* a response — during which the audio and video are recorded. These two windows are temporally offset by several seconds within the same trial. The EAV repository's preprocessing reflects this: `SELECTED_CLASSES = [1, 3, 5, 7, 9]` selects listening trials for EEG, while the vision preprocessor filters for speaking clips.

Resolution. This asymmetry cannot be corrected without re-downloading and re-preprocessing the raw EAV data (approximately 46 GB from Zenodo), which was impractical. It was instead documented as a methodological caveat for the suppression matrix interpretation: the EEG signal and the audio/video signals are both labelled with the same emotion *category* (the cue), but they are drawn from different phases of the trial. Cross-modal agreement in this context means agreement between internal neural state during *listening* and external behavioural expression during *speaking* — a meaningful but specific form of cross-modal coherence.

7.5 Methodological Challenge

C13 — Research Framing Pivot

Symptom. At approximately the project midpoint, the original research framing — "continuous in-the-wild emotion tracking" — was found to be infeasible for the Phase 1 scope. The planned demonstration of real-time emotion tracking on arbitrary video could not be delivered.

Root cause. EEG cannot be collected from in-the-wild audiovisual recordings (films, social media videos, surveillance feeds). The EEG modality requires physical contact with the subject's scalp via active electrodes. Any system that tracks emotion from arbitrary video content would therefore be operating with EEG zeroed or absent — effectively a bimodal audio-visual system, not the trimodal system the thesis had framed as its contribution.

Resolution. The research framing was pivoted from accuracy-maximisation on arbitrary video to cross-modal affective coherence characterisation on the EAV laboratory corpus. This reframing produced a stronger and more scientifically precise thesis contribution: instead of claiming to build an in-the-wild tracker that could not be validated, the thesis characterises the structure of within-trial cross-modal disagreement at population scale — a contribution that is directly grounded in the available data and directly extensible to future clinical validation. All existing code and experimental artefacts were preserved; only the analysis strategy and the framing of the headline contribution changed.

Chapter 8

Conclusion

8.1 Summary of Contributions

This thesis set out to determine whether trimodal cross-modal attention fusion could detect and characterise the structure of affective incoherence — the systematic divergence between voluntary emotional expression (audio, video) and involuntary neural response (EEG) — at population scale. Both the fusion performance objective and the coherence-characterisation objective were achieved on the EAV corpus. The `TrimodalAttentionFusion` module reached 80.2% mean test accuracy across 42 subjects, confirming that a jointly trained attention mechanism matches or exceeds naïve softmax averaging. More substantively, the 5×5 suppression matrix produced four interpretable, cross-population patterns of affective incoherence — bidirectional Anger–Happiness confusion and directional Sadness/Calmness→Neutral suppression — validated on a sample of 5,040 test trials without the EEG-Neutral inflation artefact that had appeared in the 3-subject pilot.

This thesis makes five contributions to the multimodal affective computing literature.

Contribution 1: First evaluation of cross-modal attention fusion on EAV. The EAV corpus has, since its publication in 2024, been evaluated only for per-modality baselines by its authors. This thesis provides the first multimodal fusion evaluation on EAV, establishing four reference numbers: naïve late fusion (80.0%), cross-attention fusion (80.2%), and the per-modality means against which both are compared. These numbers are reproducible from the open-source codebase at [shafinmuffinn/thesis_p2](https://github.com/shafinmuffinn/thesis_p2).

Contribution 2: Systematic identification and resolution of EAV repository bugs. Five bugs in the EAV open-source repository were identified and resolved: a forward-hook return-value defect in EEGNet (causing systematic training failure), a train/eval mode cycling defect in the shared trainer, a classifier-head mismatch in the vision trainer, a RAM-exhaustion failure in vision preprocessing, and a dependency conflict introduced by facenet-pytorch. The EEGNet bugs in particular — which caused EEG accuracy to be chance-level in the repository's published state — have implications for any researcher attempting to reproduce or extend the EAV paper's EEGNet baseline using the PyTorch path.

Contribution 3: Trimodal cross-modal attention fusion module. The `TrimodalAttentionFusion` module is an open-source, tested, and extensible trimodal fusion architecture specifically designed for the EAV modality dimensions. It satisfies two architectural commitments: (a) it returns per-modality embeddings as named dictionary outputs, enabling future temporal head attachment; (b) it produces finite predictions when EEG is zeroed, enabling inference in EEG-unavailable deployment scenarios.

Contribution 4: First suppression matrix characterisation on EAV. The 5×5 suppression matrix computed over 5,040 test trials constitutes the first population-scale characterisation of cross-modal affective coherence on EAV. Three dominant patterns are identified — bidirectional Anger–Happiness high-arousal confusion, directional Sadness→Neutral suppression, and directional Calmness→Neutral suppression — all confirmed as cross-population phenomena. This matrix provides a reference baseline against which future coherence-detection methods on EAV can be compared.

Contribution 5: Reframing of the EAV evaluation task. The thesis's central methodological contribution is the reframing of multimodal emotion recognition on EAV from a pure accuracy-maximisation problem to a coherence-characterisation problem. This reframing is motivated by the epistemological asymmetry between EEG (involuntary, internal) and audio-video (voluntary, external), and by the clinical significance of cross-modal divergence. It provides a research agenda for future EAV-based work that goes beyond "which architecture achieves the highest accuracy" to "what does cross-modal disagreement tell us about affective regulation?"

8.2 Limitations

Four limitations are explicitly acknowledged.

Limitation 1: Controlled corpus, not spontaneous data. The EAV corpus uses cue-based emotion elicitation. The "suppression events" detected in the matrix may partly reflect imperfect emotion simulation (where a participant correctly displays the cued emotion in audio-video but fails to enter the corresponding physiological state fully) rather than genuine social suppression. This confound is irreducible without cross-referencing against a concurrent internal-state measure.

Limitation 2: Within-subject evaluation only. All accuracy figures are within-subject (training and test sets from the same individual). Cross-subject evaluation (LOSO or k-fold) is the standard credibility check in affective computing and is required for any generalisation claim. The within-subject results reported here are strong but cannot be extrapolated to the cross-subject setting.

Limitation 3: Single pooled-vector feature resolution. The per-modality encoders produce single pooled vectors rather than token sequences. This limits the cross-modal information accessible to the attention fusion module and likely suppresses the attention module's potential advantage over naïve averaging. Replacing the pooled-vector encoders with sequence-producing encoders (e.g., applying LSTM or TCN to sequences of per-frame ViT representations) is the primary architectural improvement available.

Limitation 4: No clinical validation. The suppression matrix findings are validated behaviorally (cross-population consistency, stable base rate, absence of the EEG-Neutral inflation confound) but not clinically. The patterns may or may not correspond to clinically meaningful suppression in healthy participants, and the methodology has not been applied to any clinical population. Any clinical inference from the present results would be premature.

8.3 Future Work

Three workstreams are identified as Phase 2 extensions, each directly building on the architectural and methodological infrastructure developed in this thesis.

Phase 2.1: Clinical population validation. The suppression-matrix methodology, applied to a population with a known affective dysregulation diagnosis (e.g., alexithymia measured by the Toronto Alexithymia Scale), would test whether the cross-modal coherence score discriminates diagnostic groups from healthy controls. This is the primary scientific value of the methodology: the EAV-based characterisation provides a reference baseline, and deviations from that baseline in clinical populations are the signal of interest.

Phase 2.2: In-the-wild temporal emotion tracking. The architecture designed in this thesis — with per-modality embeddings exposed as named outputs — supports attachment of a temporal head (LSTM or Residual-TCN) for continuous emotion tracking on temporally-annotated video corpora such as DFEW or MAFW. This extension requires: a temporally-annotated in-the-wild corpus; a voice activity detector for audio; a robust face detector and tracker for video; and a temporal decoder consuming the per-modality embedding sequences. The per-modality encoders (AST, ViT, EEGNet) can be frozen, making this an incremental addition rather than a full redesign.

Phase 2.3: MERCL contrastive pre-training. The supervised contrastive pre-training scheme of Lee, Kim and Kim (2024) — comprising intra-modal contrastive loss (AMCL), inter-modal alignment loss (EMCL), and sample-wise alignment (SMCL) — was dropped from the present thesis under schedule pressure. Its re-introduction in Phase 2 would test whether cross-modal contrastive alignment of the per-modality encoder representations, before fine-tuning the fusion head, materially improves either classification accuracy or the signal-to-noise ratio of the suppression matrix. The hypothesis, based on the Lee et al. ablation results (−4.04% without contrastive pre-training), is that contrastive pre-training meaningfully improves cross-modal feature alignment and would manifest as a larger fusion gain on EAV. Concretely, AMCL would cluster intra-modal class representations, EMCL would align same-class representations across modalities, and SMCL would close the modality-specific embedding gap with a margin term α . All three losses operate on the pre-classifier embeddings already exposed by the `forward()` contract, requiring no architectural changes.

Closing remark. The thesis opens and closes with the same asymmetry: EEG is involuntary; audio and video are not. This asymmetry, usually a confound in fusion system design, is here the signal. The suppression matrix transforms that signal into a population-level map of affective regulation. Whether that map reflects genuine emotion suppression, simulation artefacts, or cross-modal latency mismatches remains an open question — one that the behavioral evidence in this thesis positions cleanly for future clinical investigation.

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